



T.C.

FATİH SULTAN MEHMET VAKIF UNIVERSITY

Faculty of Engineering

Department of Software Engineering

Software Engineering Design II

Courses Project

Ricci Flow-Like Dynamics in DNNs

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Project Supervisor: Assoc. Prof. Burcu Bektaş Demirci

Istanbul, [June 2026]



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Istanbul, [June 2026]

Preface

This report presents the final outcomes of the project titled Ricci Flow-Like Dynamics in DNNs. The study investigates how deep neural networks transform data geometry across successive layers and whether these transformations can be interpreted through Ricci flow-like dynamics. The main purpose of the project is to construct a reproducible analysis pipeline that represents layer activations as k -nearest-neighbor graphs, computes discrete Ricci curvature and graph geodesic quantities, and evaluates the relationship between curvature-driven geometric change and classification performance.

The scope of the work includes global Ricci coefficient analysis, local and layer-wise Ricci measurements, and experimental evaluations on synthetic data, MNIST, Fashion-MNIST, and CIFAR-10 binary classification tasks. In addition to reproducing the core geometric-flow interpretation, the project explores Ricci-based early stopping, architecture screening, and the limitations of using early geometric signals as performance predictors.

We would like to express our sincere gratitude to Ass. Soc. Prof. Burcu Bektaş Demirci, Ass. Prof. Zeki Kuş, and Ass. Soc. Prof. Ali Nizam for their valuable guidance, support, and constructive feedback throughout the course, which significantly contributed to the successful completion of this study.

12.06.2026

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Abbreviations

DNN	Deep Neural Network
k-NN	k-Nearest Neighbor
MNIST	Modified National Institute of Standards and Technology
fMNIST	Fashion-MNIST
CIFAR-10	Canadian Institute for Advanced Research 10-class dataset
ReLU	Rectified Linear Unit
RMSprop	Root Mean Square Propagation
AF	Augmented Forman-Ricci
OR	Ollivier-Ricci
ES	Early Stopping

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List of Symbols

V	Vertex set of the layer-wise k -NN graph
E_k^l	Edge set of the k -NN graph at layer l
g_l	Total graph distance at layer l
$\gamma_k^l(i, j)$	Global shortest-path distance between nodes i and j
$d_l(x, y)$	Local graph distance between points x and y at layer l
η_l	Change in total graph distance between layers
$\eta_l(x)$	Average local distance change around point x
$R_k^l(i, j)$	Forman-Ricci curvature of edge (i, j)
$\mathcal{AF}_l(x, y)$	Augmented Forman-Ricci curvature of edge (x, y)
Ric_l	Total Ricci curvature at layer l
$\mathcal{O}(x, y)$	Ollivier-Ricci curvature of edge (x, y)
$\mathcal{O}_l(x)$	Local scalar curvature around point x at layer l
ρ_k	Global Ricci coefficient
z_k	Fisher-normalized global Ricci coefficient
$\rho(l)$	Layer Ricci coefficient
$\bar{\eta}, \bar{\text{Ric}}$	Layer-wise averages used in the global coefficient
$\bar{\eta}_l, \bar{\mathcal{O}}_l$	Point-wise averages used in the layer coefficient
W_1	1-Wasserstein distance

RICCI FLOW-LIKE DYNAMICS IN DNNS

SUMMARY

This study investigates deep neural networks from a geometric perspective and focuses on the idea that learning can be interpreted as a sequence of Ricci flow-like transformations. Deep neural networks are highly effective in classification and representation learning, yet the internal organization of data across hidden layers remains difficult to interpret. A trained model does not simply map inputs to outputs in a single step; instead, it repeatedly transforms the input data through a chain of linear mappings and nonlinear activation functions. These transformations change the relative positions, distances, neighborhoods, and class structures of the data. The central motivation of this report is to make these hidden geometric changes measurable.

The conceptual foundation of the work comes from geometric flow theory. In differential geometry, Ricci flow describes the evolution of a metric according to curvature. Positively curved regions tend to contract, while negatively curved regions tend to expand. This process smooths irregularities and can reveal simpler underlying structures. The analogy used in this project is that deep neural networks may perform a comparable operation on data representations. As data moves from one layer to another, samples from the same class often become more compact, while samples from different classes become more separable. If this reorganization is connected to curvature-driven changes in the representation space, then tools from discrete geometry can be used to study learning dynamics.

The methodology models each layer's activations as a point cloud and converts this point cloud into a k -nearest neighbor graph. The graph acts as a discrete approximation of the data manifold at that layer. Once the graph is constructed, shortest-path distances are used to estimate geodesic relationships between data points, and discrete Ricci curvature is used to measure the local and global geometric structure of the graph. The report considers both global and local perspectives. The global Ricci coefficient evaluates the relationship between changes in total geodesic distance and total graph curvature across layers. The local and layer-wise Ricci coefficients examine how neighborhood-level curvature and distance changes behave within individual layer transitions.

Several curvature formulations are used or discussed in the report. Basic Forman-Ricci curvature provides a computationally efficient graph-based curvature measure based mainly on node degrees. Augmented Forman-Ricci curvature extends this idea by incorporating shared-neighbor structure, which allows the graph measure to capture triangular local connectivity. Ollivier-Ricci curvature offers a transport-

based interpretation by comparing probability distributions over neighboring nodes. These curvature tools allow the project to evaluate whether the network’s internal transformations resemble a discrete curvature-driven flow.

The experimental design includes synthetic and real-world binary classification tasks. The synthetic datasets are useful because they make geometric and topological difficulty visible in low-dimensional space. Some datasets are geometrically entangled but separable, while others require more complex transformations to separate the classes. Real-world experiments use MNIST, Fashion-MNIST, and CIFAR-10 tasks. MNIST provides relatively clean handwritten digit classes, Fashion-MNIST introduces more visually similar object categories, and CIFAR-10 adds the complexity of natural RGB images. This range of datasets makes it possible to test whether Ricci-based measurements behave consistently across increasing levels of data complexity.

The architecture analysis compares narrow, wide, flat, and bottleneck neural networks at different depths. The experiments evaluate how depth and representational capacity influence both classification accuracy and Ricci flow-like behavior. In the global experiments, deeper networks generally show stronger Ricci flow-like behavior than shallower ones. Wide architectures tend to achieve the highest classification accuracy, suggesting that sufficient representational capacity helps the network perform smoother and more effective geometric transformations. Bottleneck architectures remain competitive in some settings, indicating that compression can also contribute to meaningful geometric simplification, although excessive compression may harm performance in more difficult tasks.

The report also studies local Ricci behavior across layers. This layer-wise analysis shows that geometric transformations are not uniformly distributed throughout the network. Early, middle, and late layers may play different roles in reorganizing the feature space. In several experiments, later layers produce less stable or less negative Ricci values, suggesting that output-adjacent transformations may reflect classifier compression rather than general representation organization. This observation is important because it shows that averaging Ricci values across all layers can hide meaningful depth-dependent patterns. For that reason, the report treats layer order as an essential part of the geometric interpretation.

Another part of the study examines Ricci-based early stopping. Standard early stopping usually relies on validation loss or validation accuracy. The Ricci-based alternative asks whether curvature stabilization can identify meaningful training transitions earlier. The experiments show that Ricci-based stopping often triggers earlier than standard validation-based stopping. This suggests that curvature dynamics can capture changes in representation geometry before they become fully visible through accuracy curves. However, the early signal is sometimes too aggressive and can stop training before the model reaches its best performance, especially in bottleneck architectures. Therefore, Ricci-based early stopping is presented as a complementary diagnostic signal rather than a standalone replacement for standard early stopping.

The predictive Ricci experiments explore whether early curvature behavior can forecast long-term model performance. On Fashion-MNIST, early local Ricci scores show a statistically significant but moderate relationship with long-term accuracy. This means that early geometric flow contains some useful information about future performance. Some models with strong early Ricci scores do not become the best

long-term performers, and some high-performing models do not have the strongest early Ricci values. The signal is therefore useful for coarse filtering but not reliable enough for precise architecture ranking.

On CIFAR-10, the local Ricci signal becomes weaker and statistically unreliable. Extending the Ricci observation window and testing multiple neighborhood sizes does not fully solve this issue. The increased complexity of natural image data appears to make early local geometry noisier. A later comparison between local Ricci, raw global Ricci, and Fisher-normalized global Ricci shows that global measurements may reveal different relationships from local measurements. In particular, depth-normalized global Ricci behavior on CIFAR-10 displays an inverse relationship with long-term accuracy. This suggests that strong early contraction is not always beneficial in complex tasks and may sometimes indicate premature or overly rigid representation formation.

Overall, the study supports the view that deep neural networks perform meaningful geometric transformations that can be studied using discrete Ricci curvature. The findings show that Ricci-based measurements are valuable for interpretation, architecture comparison, and analysis of training dynamics. At the same time, the results also show clear limitations. Curvature signals depend on dataset complexity, architecture type, graph construction scale, and whether the analysis is local or global. For this reason, Ricci curvature should be understood as a complementary geometric lens rather than a universal predictor of model quality.

The contribution of this project is a structured software-oriented framework for measuring Ricci flow-like behavior in neural networks. By combining layer activations, k-nearest neighbor graphs, geodesic distances, and discrete curvature measures, the report provides a practical method for observing how learned representations evolve. This framework can support future work on geometry-aware diagnostics, architecture screening, training control, and interpretability in deep learning systems.

DERİN SİNİR AĞLARINDA RICCI AKIŞINA BENZER DİNAMİKLER

ÖZET

Bu çalışma, derin sinir ağlarının öğrenme sürecinde veriyi katmanlar boyunca nasıl dönüştürdüğünü geometrik bir bakış açısıyla incelemektedir. Derin öğrenme modelleri günümüzde görüntü sınıflandırma, örüntü tanıma ve temsil öğrenme gibi birçok alanda yüksek başarımlar elde etmektedir. Bununla birlikte, bu modellerin iç katmanlarında verinin nasıl yeniden düzenlendiği, sınıfların hangi ara dönüşümler sonucunda ayrıştığı ve öğrenmenin temsil uzayında ne tür geometrik izler bıraktığı hâlâ açıklanması güç bir problemdir. Bir derin sinir ağı, girdiyi doğrudan sonuca dönüştüren tek adımlı bir yapı değildir; bunun yerine veri, ardışık doğrusal dönüşümler ve doğrusal olmayan aktivasyonlar aracılığıyla çok sayıda ara temsilden geçer. Bu süreçte örnekler arasındaki uzaklıklar, komşuluk ilişkileri, sınıf kümelenmeleri ve temsil uzayının genel geometrisi değişir. Çalışmanın temel amacı, bu iç dönüşümleri yalnızca doğruluk ve kayıp değerleriyle değil, ayrık diferansiyel geometri araçlarıyla da ölçülebilir hale getirmektir.

Çalışmanın kuramsal çıkış noktası geometrik akış kavramıdır. Diferansiyel geometride Ricci akışı, bir uzayın metriğinin eğriliğe bağlı olarak evrilmesini tanımlar. Pozitif eğriliğe sahip bölgeler daralma eğilimi gösterirken, negatif eğriliğe sahip bölgeler genişleyebilir. Bu süreç, düzensiz geometrik yapıları daha düzenli hale getirerek altta yatan daha basit biçimleri ortaya çıkarabilir. Derin sinir ağları için kullanılan benzetme de bu noktada ortaya çıkar. Eğitim görmüş bir ağda aynı sınıfa ait örneklerin giderek birbirine yaklaşması, farklı sınıflara ait örneklerin ise ayrışması, temsil uzayında bir tür geometrik sadeleşme gerçekleştiğini düşündürmektedir. Eğer bu sadeleşme eğrilik ve uzaklık değişimleriyle ilişkilendirilebilirse, öğrenme süreci Ricci akışına benzer ayrık bir geometrik dönüşüm olarak yorumlanabilir.

Bu amaçla çalışmada her katmandaki aktivasyon çıktıları birer nokta bulutu olarak ele alınmıştır. Bu nokta bulutları üzerinde k -en yakın komşu grafikleri kurulmuş ve her katmandaki veri temsili ayrık bir manifold yaklaşımıyla modellenmiştir. Graf yapısı, veri örnekleri arasındaki yerel komşuluk ilişkilerini açık hale getirirken, en kısa yol uzaklıkları da bu graf üzerinde jeodezik uzaklıkların hesaplanmasına imkân vermektedir. Böylece her katmandaki temsil uzayının yalnızca Öklidyen koordinatları değil, aynı zamanda graf tabanlı iç uzaklık yapısı da analiz edilebilmektedir. Bu yapı üzerine ayrık Ricci eğriliği ölçütleri uygulanarak ağı katmandan katmana verinin geometrisini nasıl değiştirdiği incelenmiştir.

Çalışmada hem global hem de yerel geometrik ölçütler kullanılmıştır. Global Ricci katsayısı, katmanlar boyunca toplam jeodezik uzaklıktaki değişim ile toplam eğrilik arasındaki ilişkiyi ölçer. Bu katsayı, ağıın genel temsil geometrisinin Ricci akışına benzer biçimde evrilip evrilmediğini değerlendirmek için kullanılır. Yerel ve katman bazlı Ricci katsayıları ise her bir katman geçişinde komşuluk düzeyindeki uzaklık ve eğrilik değişimlerini inceler. Bu yerel yaklaşım, ağıın tamamını tek bir özet katsayıya indirgemek yerine, farklı katmanların öğrenme sürecindeki rollerini daha ayrıntılı biçimde gözlemlemeyi sağlar. Böylece erken, orta ve geç katmanların temsil geometrisini farklı biçimlerde etkileyip etkilemediği analiz edilebilir.

Eğrilik ölçümü için temel Forman-Ricci eğriliği, artırılmış Forman-Ricci eğriliği ve yaklaşık Ollivier-Ricci eğriliği gibi ayrıık yöntemler ele alınmıştır. Forman-Ricci eğriliği graf üzerindeki kenarların yerel bağlantı yapısına dayalı, hesaplama açısından verimli bir ölçüt sunar. Artırılmış Forman-Ricci eğriliği ise ortak komşuları ve üçlü çevrimleri dikkate alarak yerel bağlantı desenlerini daha zengin biçimde yakalar. Ollivier-Ricci eğriliği, komşuluk dağılımları arasındaki taşıma uzaklığına dayandığı için daha farklı ve olasılıksal bir geometrik yorum sağlar. Bu ölçütler birlikte kullanıldığında, derin sinir ağlarının iç dönüşümlerinin yalnızca sınıflandırma sonucuna değil, aynı zamanda temsil uzayındaki geometrik akışa göre de değerlendirilebilmesi mümkün olur.

Deneyisel çalışma farklı zorluk seviyelerine sahip sentetik ve gerçek dünya veri kümeleri üzerinde yürütülmüştür. Sentetik veri kümeleri, sınıfların geometrik olarak iç içe geçtiği veya topolojik olarak ayrışmasının zor olduğu durumları görselleştirmek açısından önemlidir. Bu veri kümeleri sayesinde ağıın basit doğrusal ayrıımlardan daha karmaşık manifold dönüşümlerine kadar farklı problemler karşısındaki davranışı gözlenebilmiştir. Gerçek dünya deneylerinde MNIST, Fashion-MNIST ve CIFAR-10 veri kümeleri kullanılmıştır. MNIST görece temiz ve iyi ayrışabilen el yazısı rakam sınıfları sağlarken, Fashion-MNIST görsel olarak birbirine daha yakın giyim nesnelerini içerir. CIFAR-10 ise doğal renkli görüntülerden oluştuğu için daha karmaşık, gürültülü ve yüksek boyutlu bir sınıflandırma ortamı sunar.

Mimari karşılaştırmalar kapsamında dar, geniş, düz ve darboğaz yapılı derin sinir ağları incelenmiştir. Dar mimariler daha sınırlı temsil kapasitesi sunarken, geniş mimariler aynı katman sayısında daha fazla nöron içerdiği için veriyi dönüştürme gücü bakımından daha zengindir. Düz mimariler tüm gizli katmanlarda sabit genişlik kullanırken, darboğaz mimarileri bilgiyi belirli katmanlarda sıkıştırarak yeniden genişletir. Bu farklı yapıların karşılaştırılması, derinlik, genişlik ve bilgi sıkıştırmanın Ricci akışına benzer geometrik davranış üzerindeki etkisini anlamak açısından önemlidir. Deneyler, daha derin ve daha yüksek kapasiteli bazı ağların daha belirgin geometrik akış davranışı gösterebildiğini ortaya koymuştur.

Global Ricci tabanlı deneylerde, özellikle gerçek dünya veri kümelerinde, derin ağların daha güçlü Ricci akışına benzer davranış sergilediği gözlemlenmiştir. Beş katmanlı ve on bir katmanlı mimariler karşılaştırıldığında, daha derin ağlarda geometrik ölçütler ile sınıflandırma doğruluğu arasındaki ilişkinin daha belirgin hale geldiği görülmüştür. Geniş mimariler genel olarak en yüksek doğruluk değerlerine ulaşmış, bu da temsil kapasitesinin geometrik sadeleşme açısından önemli olduğunu göstermiştir. Bununla birlikte darboğaz mimarilerinin de bazı görevlerde rekabetçi performans üretmesi, kontrollü bilgi sıkıştırmanın veri manifoldunu daha düzenli hale getirmeye katkı sağlayabileceğini düşündürmektedir.

Yerel Ricci tabanlı analizler, geometrik dönüşümlerin ağ boyunca eşit dağılmadığını göstermiştir. Bazı katmanlar veri temsili daha güçlü biçimde düzenlerken, bazı katmanlarda Ricci katsayıları daha kararsız veya daha az negatif değerlere yaklaşabilmektedir. Özellikle son katmanlara yakın bölgelerde görülen bu davranış, çıkış sınıflandırıcısına doğru gerçekleşen sıkıştırmanın genel temsil geometrisinden farklı bir karakter taşıyabileceğini göstermektedir. Bu nedenle çalışmada katman sırasının korunması önemli bir ilke olarak ele alınmıştır. Ricci katsayılarını yalnızca tek bir ortalamaya indirgemek, hangi katmanda hangi geometrik değişimin gerçekleştiğine dair önemli bilgileri gizleyebilir.

Çalışmanın bir diğer önemli yönü, Ricci tabanlı ölçütlerin eğitim sürecinde erken durdurma sinyali olarak kullanılıp kullanılamayacağını araştırmasıdır. Geleneksel erken durdurma yöntemleri genellikle doğrulama kaybı veya doğrulama doğruluğundaki iyileşmenin durmasına dayanır. Ricci tabanlı yaklaşım ise temsil geometrisindeki eğrilik değişimlerinin daha erken bir yakınsama veya kararlılık sinyali verip vermediğini inceler. Deneylerde Ricci tabanlı erken durdurma ölçütünün çoğu durumda standart erken durdurmadan daha erken tetiklendiği görülmüştür. Bu bulgu, geometrik sinyalin eğitim dinamiklerindeki bazı değişimleri doğruluk eğrilerinden önce yakalayabildiğini düşündürmektedir.

Ancak Ricci tabanlı erken durdurma sinyalinin tek başına güvenilir bir karar mekanizması olmadığı da görülmüştür. Bazı mimarilerde, özellikle darboğaz yapıları ve bilgi sıkıştırmanın güçlü olduğu ağlarda, Ricci sinyali çok erken tetiklenerek modelin nihai başarımına ulaşmadan eğitimin durdurulmasına neden olabilecek bir eğilim göstermiştir. Bu nedenle çalışma, Ricci tabanlı erken durdurmaya standart doğrulama temelli yöntemlerin yerine geçecek bir araç olarak değil, onları tamamlayabilecek bir geometrik tanı sinyali olarak değerlendirmektedir. Bu yaklaşım, eğitim sırasında modelin yalnızca çıktı performansını değil, temsil uzayındaki yapısal değişimini de takip etmeyi mümkün kılar.

Mimari tarama ve uzun dönem performans tahmini deneyleri, erken dönem Ricci davranışının ne ölçüde öngörü gücü taşıdığını araştırmıştır. Fashion-MNIST üzerinde yürütülen deneylerde, ilk eğitim dönemlerinde hesaplanan yerel Ricci skorlarının uzun dönem doğrulukla istatistiksel olarak anlamlı fakat orta düzeyde bir ilişki gösterdiği bulunmuştur. Bu sonuç, erken geometrik akış sinyalinin mimariler hakkında bazı yararlı bilgiler taşıdığını göstermektedir. Bununla birlikte ilişki güçlü değildir; erken Ricci skoru yüksek olan her mimari uzun vadede en iyi performansı göstermemekte, yüksek doğruluğa ulaşan bazı mimariler de en güçlü erken Ricci sinyaline sahip olmamaktadır. Bu nedenle erken Ricci ölçütleri kesin bir sıralama aracı olmaktan çok, kaba bir mimari filtreleme sinyali olarak yorumlanmalıdır.

CIFAR-10 üzerinde yapılan deneyler, veri karmaşıklığı arttıkça yerel Ricci sinyalinin daha gürültülü hale geldiğini göstermiştir. Kırk epokluk daha uzun bir gözlem penceresi kullanılmasına ve farklı k-en yakın komşu değerleri denenmesine rağmen, yerel Ricci skorları ile uzun dönem doğruluk arasında güçlü ve güvenilir bir ilişki kurulamamıştır. Bu durum, doğal görüntü verilerinde erken dönem yerel manifold geometrisinin sınıflandırma başarısını açıklamakta yetersiz kalabileceğini göstermektedir. CIFAR-10 üzerinde ayrıca yerel Ricci, ham global Ricci ve Fisher dönüşümüyle derinlik-normalize edilmiş global Ricci ölçütleri karşılaştırılmıştır. Bu karşılaştırma, global ölçütlerin yerel ölçütlerden farklı eğilimler ortaya çıkarabileceğini göstermiştir.

Bu bulgular, Ricci tabanlı geometrik ölçümlerin veri kümesine, mimariye, komşuluk ölçeğine ve ölçümün yerel ya da global olmasına duyarlı olduğunu ortaya koymaktadır. Daha basit veya daha düzenli veri kümelerinde erken geometrik sinyaller model performansı hakkında daha anlamlı bilgiler verebilirken, daha karmaşık görsel sınıflandırma problemlerinde aynı sinyal daha zayıf kalabilmektedir. Bu nedenle Ricci eğriliği, derin öğrenme modellerinin başarımını tek başına açıklayan evrensel bir ölçüt olarak değil, model yorumlanabilirliğini artıran ve eğitim dinamiklerini farklı bir açıdan görünür kılan tamamlayıcı bir analiz aracı olarak ele alınmalıdır.

Sonuç olarak bu çalışma, derin sinir ağlarının öğrenme davranışını temsil uzayındaki geometrik dönüşümler üzerinden inceleyen yazılım odaklı bir analiz çerçevesi sunmaktadır. Katman aktivasyonlarının k-en yakın komşu grafikleriyle modellenmesi, en kısa yol uzaklıklarının hesaplanması ve ayrık Ricci eğriliği ölçütlerinin uygulanması sayesinde, ağın veriyi katmanlar boyunca nasıl düzenlediği ölçülebilir hale getirilmiştir. Elde edilen sonuçlar, derin sinir ağlarının birçok durumda Ricci akışına benzer bir geometrik sadeleşme davranışı sergilediğini göstermektedir. Aynı zamanda bu davranışın her veri kümesi ve her mimari için aynı biçimde ortaya çıkmadığı, özellikle karmaşık doğal görüntü görevlerinde daha dikkatli yorumlanması gerektiği görülmüştür.

Bu proje, derin öğrenme sistemlerinin yalnızca çıktı doğruluğu üzerinden değil, iç temsil geometrisi üzerinden de değerlendirilebilir olduğunu vurgulamaktadır. Ayrık Ricci eğriliğine dayalı yaklaşım; model yorumlanabilirliği, mimari karşılaştırma, erken eğitim dinamiklerinin izlenmesi ve geometri-duyarlı tanı araçlarının geliştirilmesi için yararlı bir temel sunmaktadır. Gelecekte bu çerçeve farklı veri kümeleri, daha büyük ağ yapıları, evrimsel ve dönüştürücü tabanlı mimariler ile daha gelişmiş eğrilik ölçütleri üzerinde genişletilebilir. Böylece derin öğrenmenin iç işleyişini açıklamaya yönelik geometrik yaklaşımlar hem kuramsal hem de pratik açıdan daha güçlü hale getirilebilir.

1. INTRODUCTION

This graduation project investigates deep learning through the lens of geometric flows, with a particular emphasis on Ricci flow as a framework for interpreting layer-wise transformations in neural networks. At the core of deep learning models are deep neural networks (DNNs), which transform data step by step through multiple layers to find meaningful patterns. Although these networks are widely used, what exactly happens to the data as it passes through each layer is still not fully clear. In this context, understanding this internal process can help us see how deep networks actually learn and simplify complex information.

Ricci flow is a process that gradually smooths the curvature of a shape, just like heat diffuses across a surface, making it more uniform over time. A recent paper by Baptista et al. [1] draws this connection directly, showing that each layer of a neural network can be viewed as a discrete step of Ricci flow. They build k-nearest neighbor (k-NN) graphs at every layer to represent how data points relate to each other, and they calculate curvature values to track how these relationships change. When curvature and distance evolve in opposite directions, as expected under Ricci flow, it suggests that the network is smoothing the data geometry as it learns. In this study, we look at deep learning from a geometric perspective, inspired by this connection between Ricci flow and layer-wise representation learning.

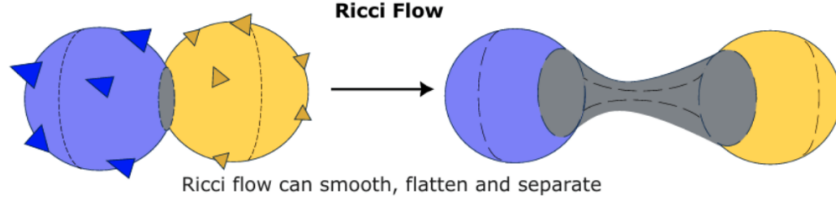


Figure 1.1. Conceptual behavior of Ricci flow: an irregularly shaped manifold with two bulging regions connected by a thin tube evolves under Ricci flow, with positively curved bulges contracting and negatively curved tubes expanding, eventually smoothing into simpler geometric forms (adapted from Baptista et al.) [1]

1.1 Geometric Flow

A geometric flow is an evolution equation that deforms a geometric object, such as a curve, surface, or manifold, over an abstract time parameter according to an intrinsic geometric quantity. In general form, a flow is written as

$$\frac{\partial g_t}{\partial t} = F(g_t), \quad (1.1)$$

where g_t denotes a time-dependent geometric structure like a metric, and F encodes the deformation rule, such as curvature. The motivation for geometric flows is twofold: first, they smooth out irregularities in a process analogous to diffusion, revealing canonical shapes; and second, they expose deep structural properties by driving a space toward simpler, often uniform, states. Classical examples include curve shortening flow, which

evolves plane curves by curvature, shrinking and circularizing them; mean curvature flow, which evolves surfaces by mean curvature, smoothing them; Willmore flow, which reduces the bending energy of surfaces; and Ricci flow, where Riemannian metrics evolve according to the Ricci curvature tensor, contracting positively curved regions and expanding negatively curved ones. Among these, Ricci flow is especially attractive for high-dimensional data analysis because it acts directly on the metric i.e., on distances, precisely the object neural networks implicitly manipulate across layers.

1.2 Motivation and Significance

In modern deep learning, we observe empirically that intermediate representations become increasingly organized: same-class points cluster; different classes separate. This qualitative “untangling” suggests a curvature-driven effect: dense, positively curved regions contract, while sparse, negatively curved regions expand. Ricci flow offers a principled mathematical analogy for this phenomenon and, when discretized on graphs, yields computable proxies, such as layerwise k-NN geodesics and discrete Ricci curvature, that turn the analogy into measurable quantities. Establishing, quantifying, and leveraging this link matters for both understanding and improving learning: it can provide geometry-aware diagnostics, and inform architectural or training decisions.

1.3 Aim of the Project

Building on the “Deep Learning as Ricci Flow” perspective, our aim is to (i) present a clear, software-oriented formulation of the layerwise geometric evolution in DNNs; (ii) reproduce and extend the discrete pipeline that measures curvature–metric interactions across layers; and (iii) convert these measurements into actionable tools for model interpretation and training control.

1.4 Scope of Problems

The scope of this project encompasses several interconnected problems:

Conceptual grounding: Provide a concise, unified treatment of Ricci flow and deep learning representations, aligning geometric notation with the practical data structures produced by DNNs, such as layer activations, k-NN graphs, and shortest-path geodesics.

Learning as flow: Make explicit the layer-to-layer correspondence between metric change and curvature, and formalize a workflow that explains the learning procedure via a Ricci-flow-like relation on discrete graphs.

Visualization: Develop clear, reproducible visualizations of layerwise geometry (e.g., per-layer k-NN connectivity, curvature distributions, and geodesic-spread trajectories) to make the flow behavior interpretable at multiple scales.

Early-stopping criterion: Propose and evaluate a geometry-based early-stopping signal using Ricci coefficients, which represent the correlations between layerwise curvature and metric change, and compare its timing and robustness against validation-loss heuristics.

1.5 Organization of the Report

The thesis proceeds as follows. First, we review core background on DNN representations, geometric flows, and the discrete graph framework used in the analysis, including k -NN graphs, geodesic distances, and discrete Ricci curvature. Next, we present the methodology for computing layerwise geometric quantities and the Ricci coefficient, including design choices for distance metrics, neighborhood size k , graph connectivity guarantees, and computational complexity. We then provide visual and quantitative analyses across datasets and architectures. In addition to the main global Ricci-based analysis, we also include an extension to local Ricci-based measurements and a set of exploratory experiments related to this perspective. We conclude with key findings, limitations, and practical recommendations.

2. REVIEW OF THE LITERATURE

Deep neural networks (DNNs) have become widely used for modeling high dimensional and nonlinear data, thanks to their ability to approximate complex relationships across multiple layers [1, 2]. Early theoretical results, such as the universal approximation theorems of Cybenko [3] and Hornik et al. [4], showed that a sufficiently wide single-layer network can approximate any continuous or Borel-measurable function. Later work demonstrated that deeper architectures can represent certain classes of functions far more efficiently than shallow ones, sometimes requiring exponentially fewer neurons [5, 6, 7]. This efficiency comes from the compositional structure of deep models, where each layer builds upon the previous one [8, 9]. Practical achievements in computer vision and pattern recognition further confirmed the advantages of increased depth [10, 11].

Even with these foundations, theory still lags behind the remarkable empirical performance of deep learning [11]. One direction that attempts to close this gap studies the expressivity of networks with piecewise-linear activations like ReLU. Results by Montúfar et al. [5] and Pascanu et al. [8] showed that deeper networks can generate exponentially more linear regions in their decision surfaces. However, Hanin and Rolnick [12] later observed that trained networks typically use only a fraction of this potential, meaning theoretical expressivity alone does not fully explain how deep learning works in practice.

Another major line of research looks at how the geometry and topology of data change as it moves through the layers of a network. Under the manifold hypothesis, real-world data are assumed to lie on low-dimensional manifolds inside a high-dimensional space. Several studies have shown that trained DNNs tend to make these manifolds simpler and more separable as depth increases [13, 14, 15, 16, 17, 18]. For instance, Naitzat et al. [14] demonstrated that a feedforward network can take two highly entangled manifolds and transform them into two nearly spherical, linearly separable clusters at the output layer. Other work using computational topology shows that this transformation is not always smooth or monotonic, some layers may temporarily complicate the structure before it improves again [17]. Research on intrinsic dimension complements this by showing that deeper layers often reduce the effective dimensionality of the data representations [16]. Overall, manifold simplification appears strongly connected to better classification performance [17, 18].

These geometric observations have inspired attempts to analyze deep networks using continuous differential geometry. Hauser and Ray [19] proposed a Riemannian viewpoint where each layer acts as a smooth coordinate transformation. However, this approach assumes smoothness, which does not hold for common activation functions like ReLU [20]. Related perspectives interpret deep networks as discretized dynamical systems or differential equations, offering insights into stability and architectural design [21, 22, 23], but they face similar limitations due to the non-smooth, discrete nature of real networks. Because of these limitations, researchers have turned toward discrete geometric tools, especially discrete analogues of Ricci curvature. Forman Ricci curvature [27] and Ollivier Ricci curvature [28] provide ways to measure structural properties directly on graphs and point clouds. These tools have been applied to tasks such as detecting changes in evolving networks [29], describing structural features of complex graphs

[30], analyzing the influence of nodes [31], and performing community detection using discrete Ricci flow [32]. In graph neural networks, curvature has helped identify bottlenecks and over-squashing behavior, showing that geometric analysis can reveal meaningful properties of discrete learning systems [33].

A central motivation for using Ricci curvature arises from its importance in classical geometry. Hamilton introduced Ricci flow as a process that smooths a manifold’s metric by shrinking positively curved regions and expanding those with negative curvature [24]. Ricci flow became especially prominent after Perelman used Ricci flow with carefully controlled topological modifications to prove the Poincaré conjecture, a landmark result that showed how curvature-driven evolution can uncover a manifold’s topological structure [25, 26]. This connection illustrates how geometric flows can simplify complex shapes, providing the conceptual backdrop for adapting Ricci flow ideas to data analysis and machine learning.

Building on this broader geometric perspective, Baptista et al. [1] introduce a new way to connect deep learning with discrete Ricci flow. Their approach treats each layer’s representation of the data as a k -nearest-neighbor graph, computes Forman–Ricci curvature on its edges, and then examines how distances and curvature evolve across layers. From this, they define a global Ricci network flow and a corresponding Ricci coefficient that measures how strongly the DNN’s transformations resemble a Ricci flow. Across several datasets and architectures, they find that DNNs often show Ricci flow-like behavior, and importantly that networks with stronger flow-like characteristics tend to achieve higher accuracy. This provides a practical and computationally direct way to link geometric evolution to deep learning performance. Their work fills an important gap in the literature and serves as the foundation for the present study’s direction.

A more recent study [34] continues this line of work by examining neural feature geometry at a more local scale through discrete Ricci flow on graphs. Rather than considering only a global Ricci-based summary, it introduces local Ricci evolution coefficients to study how neighborhood-level geometric changes develop across successive layers. In this setting, Ollivier–Ricci curvature is used as a transport-based measure of local graph geometry, while Forman–Ricci curvature provides a computationally simpler combinatorial alternative. The study also considers augmented Forman–Ricci curvature, which incorporates shared-neighbor structure and therefore captures local connectivity more effectively than the basic Forman definition [34]. In this way, the geometric analysis is extended from a global layerwise view to a more detailed account of local changes within the learned feature space.

3. PROJECT REQUIREMENTS

3.1 Software Requirements

This project requires a computational environment capable of training and analysing deep neural networks and extracting intermediate layer representations. Python was selected as the primary programming language due to its extensive ecosystem for deep learning, numerical computation, and scientific analysis. The experiments were implemented with Python 3.10, TensorFlow/Keras 2.9 for the baseline global experiments, PyTorch 2.x for the local Ricci experiments, scikit-learn 1.x for preprocessing and k-nearest-neighbor graph construction, and Matplotlib 3.x for plots and visual outputs.

Deep neural network models were implemented using established deep learning frameworks that support flexible network architectures and access to internal layer outputs. Additional scientific computing libraries were employed to construct graph-based representations of data and to perform numerical operations required for curvature and distance calculations. The generated plot outputs include layer-wise graph visualizations, curvature trajectories, Ricci coefficient curves, early-stopping comparison plots, and architecture-screening scatter plots.

3.2 Computational Requirements

The training and analysis of deep neural networks involve computationally intensive operations, particularly when processing high-dimensional data and multiple network architectures. Therefore, hardware acceleration through the use of a GPU was required to ensure feasible training times.

The majority of matrix operations and model training procedures were executed on the GPU. Sufficient system memory was also required to store model parameters, intermediate layer outputs, and graph representations derived from test data.

3.3 Academic and Methodological Requirements

The project is grounded in prior research on the geometric and topological interpretation of deep learning models. The methodological design was informed by studies investigating how deep neural networks transform data manifolds across successive layers.

The choice to analyse deep neural networks through discrete Ricci curvature and Ricci flow-inspired metrics was motivated by the need for a global, quantitative measure of geometric transformation that is independent of network depth and width. Existing performance metrics such as classification accuracy were considered insufficient to capture these geometric properties.

Accordingly, the project required the integration of concepts from differential geometry, graph theory, and deep learning to establish a rigorous and reproducible framework for evaluating the relationship between network behaviour and classification performance.

4. SYSTEM DESIGN

This section describes the data sources, and methods used in the project. The theoretical concepts, tools, and procedures followed to achieve the objectives are explained in detail, primarily adhering to the framework of Baptista et al. [1], while incorporating a more refined geometric characterization through augmented curvature notions and layer-wise analysis as introduced by Hehl et al. [34].

This study is designed to quantify the geometric transformations a Deep Neural Network (DNN) applies to data throughout its layers during a binary classification task. We model the data at each layer as a discrete manifold using k -nearest neighbor (k -NN) graphs [1]. We then apply a form of discrete Ricci curvature to quantify the geometry of these graphs. While the foundational framework utilizes Forman-Ricci curvature [27], our implementation specifically employs the augmented Forman-Ricci curvature [34], which provides a more detailed geometric signal by incorporating contributions from three-cycles in the graph.

This approach allows us to track the geometric evolution layer by layer. To measure the strength of this evolution's resemblance to a geometric flow, we utilize two distinct but complementary metrics:

1. **Global Ricci Coefficient:** Following Baptista et al. [1], we define a "Ricci Coefficient" derived from the Pearson correlation between changes in the total manifold volume (geodesic distance) and its total curvature across all layers. This allows us to test the hypothesis that the overall geometric simplification performed by the DNN is analogous to a global Ricci flow.
2. **Layer Ricci Coefficient:** To capture the inherently local nature of these transformations, we extend the analysis using the layer Ricci coefficient introduced by Hehl et al. [34]. This metric evaluates whether the geometric transformations induced by an individual layer l align with those expected under curvature-driven dynamics, providing a more granular view of the learning process.

By integrating these global and layer-wise perspectives, the system provides a robust framework to verify if the geometric disentanglement of classes correlates with classification performance and to identify critical points such as optimal network depth or early-stopping signals [34].

4.1 Derivation of the Ricci Coefficient and Criteria

4.1.1 Continuous Ricci flow

On a smooth Riemannian manifold M , Ricci flow evolves a time-dependent metric $g(t)$ according to the curvature tensor of that metric. The standard continuous partial differential equation is

$$\begin{cases} \frac{\partial}{\partial t} g(t) = -2 \text{Ric}(g(t)), \\ g(0) = g_0, \end{cases} \quad (4.1)$$

where g_0 is the initial metric and $\text{Ric}(g(t))$ is the Ricci curvature associated with the metric at time t . The constant -2 is conventional; equivalently, one may write $\partial_t g(t) = -\alpha \text{Ric}(g(t))$ for a positive scalar α . This continuous equation cannot be applied directly to a DNN because neural networks evolve data through discrete layers and because the observed data are finite samples rather than a complete smooth manifold.

4.1.2 Discretization through layer graphs

For a trained network, let X_{test}^l denote the test-set activations at hidden layer l . Each X_{test}^l is treated as a finite sample from the feature manifold at that layer. We construct a k -nearest-neighbor graph

$$\mathcal{G}_k(X_{\text{test}}^l) = (V, E_k^l), \quad (4.2)$$

where the vertex set V corresponds to the fixed test samples and E_k^l contains the neighborhood edges induced by the activations at layer l . The parameter k is the number of nearest neighbors used to construct the graph; it determines the neighborhood resolution at which the feature geometry is observed. It must be large enough to keep the graph connected when global all-pairs geodesic distances are computed.

The graph shortest-path distance between vertices $i, j \in V$ is denoted by

$$\gamma_k^l(i, j) = d_{\mathcal{G}_k(X_{\text{test}}^l)}(i, j). \quad (4.3)$$

This quantity is the discrete analogue of a geodesic distance on the sampled feature manifold.

4.1.3 Global distance and curvature

Following Baptista et al. [1], the total graph distance at layer l is defined as

$$g_l = \sum_{i, j \in V} \gamma_k^l(i, j). \quad (4.4)$$

The change in total distance between two consecutive layers is then

$$\eta_l = g_{l+1} - g_l. \quad (4.5)$$

For the global analysis, the Forman-Ricci curvature of an unweighted edge $(i, j) \in E_k^l$ is

$$R_k^l(i, j) = 4 - \deg_k^l(i) - \deg_k^l(j), \quad (4.6)$$

where $\deg_k^l(i)$ is the degree of vertex i in $\mathcal{G}_k(X_{\text{test}}^l)$. The total curvature at layer l is

$$\text{Ric}_l = \sum_{(i, j) \in E_k^l} R_k^l(i, j). \quad (4.7)$$

The discrete analogue of the continuous Ricci flow equation is therefore written as

$$g_{l+1} - g_l = \eta_l \approx -\alpha \text{Ric}_l. \quad (4.8)$$

This relation states that the change in the graph metric between layer l and layer $l + 1$ should be negatively associated with the curvature measured at layer l .

4.1.4 Global Ricci coefficient

The global Ricci coefficient is computed as the Pearson correlation between the sequence of distance changes η_l and the sequence of total curvatures Ric_l :

$$\rho_k = \frac{\sum_{l=1}^{L-1} (\eta_l - \bar{\eta})(\text{Ric}_l - \bar{\text{Ric}})}{\sqrt{\sum_{l=1}^{L-1} (\eta_l - \bar{\eta})^2} \sqrt{\sum_{l=1}^{L-1} (\text{Ric}_l - \bar{\text{Ric}})^2}}. \quad (4.9)$$

Here, the averages used in the correlation are

$$\bar{\eta} = \frac{1}{L-1} \sum_{l=1}^{L-1} \eta_l, \quad (4.10)$$

and

$$\bar{\text{Ric}} = \frac{1}{L-1} \sum_{l=1}^{L-1} \text{Ric}_l. \quad (4.11)$$

Negative values of ρ_k indicate that total distance changes and total curvature evolve in opposite directions, which is the expected signature of global Ricci network flow-like behavior.

4.1.5 Local and layer-wise Ricci coefficients

The local analysis follows the layer-wise framework of Hehl et al. [34]. $d_l(x, y)$ denotes the graph distance between points x and y at layer l , while $O_l(x)$ denotes the local scalar curvature around point x . In our implementation, $d_l(x, y)$ is computed as a shortest-path distance on the k -nearest-neighbor graph, and $O_l(x)$ may be instantiated using approximate Ollivier-Ricci curvature or augmented Forman-Ricci curvature.

For a point x at layer l , the neighborhood $N_l(x)$ is the set of points directly connected to x by an edge in $\mathcal{G}_k(X_{\text{test}}^l)$:

$$N_l(x) = \{y \in V : (x, y) \in E_k^l\}. \quad (4.12)$$

The average change in distances from x to its immediate graph neighbors is

$$\eta_l(x) = \frac{1}{\deg_l(x)} \sum_{y \in N_l(x)} (d_{l+1}(x, y) - d_l(x, y)). \quad (4.13)$$

This quantity measures whether the local neighborhood around x expands or contracts during the transition from layer l to layer $l + 1$.

The local scalar curvature $O_l(x)$ is obtained by averaging the curvature of incident edges. When augmented Forman-Ricci curvature is used, the edge curvature is

$$\mathcal{AF}_l(x, y) = 4 - \deg_l(x) - \deg_l(y) + 3|N_l(x) \cap N_l(y)|. \quad (4.14)$$

The last term counts shared neighbors of x and y , so it incorporates triangular local connectivity into the curvature value. The corresponding local scalar curvature is

$$\mathcal{O}_l(x) = \frac{1}{\deg_l(x)} \sum_{y \in N_l(x)} \mathcal{AF}_l(x, y). \quad (4.15)$$

When Ollivier-Ricci curvature is used instead, the curvature of edge (x, y) is defined by

$$\mathcal{O}(x, y) = 1 - \frac{W_1(\mu_x, \mu_y)}{d_l(x, y)}, \quad (4.16)$$

where μ_x and μ_y are probability measures over the neighborhoods of x and y , and W_1 is the 1-Wasserstein distance. Under this choice, $\mathcal{O}_l(x)$ is obtained by averaging $\mathcal{O}(x, y)$ over $y \in N_l(x)$.

To assess whether a specific layer transition follows curvature-driven dynamics, we compute the layer Ricci coefficient

$$\rho(l) = \frac{\sum_{x \in V} (\eta_l(x) - \bar{\eta}_l)(\mathcal{O}_l(x) - \bar{\mathcal{O}}_l)}{\sqrt{\sum_{x \in V} (\eta_l(x) - \bar{\eta}_l)^2} \sqrt{\sum_{x \in V} (\mathcal{O}_l(x) - \bar{\mathcal{O}}_l)^2}}. \quad (4.17)$$

The layer-wise averages are

$$\bar{\eta}_l = \frac{1}{|V|} \sum_{x \in V} \eta_l(x), \quad \bar{\mathcal{O}}_l = \frac{1}{|V|} \sum_{x \in V} \mathcal{O}_l(x). \quad (4.18)$$

Negative values of $\rho(l)$ indicate that points with higher local curvature tend to experience local contraction while points with lower curvature tend to experience local expansion, matching the qualitative behavior expected from Ricci flow.

4.2 DNN As a Geometric Flow Generator

The Deep Neural Network (DNN) is the system responsible for generating the geometric transformations we analyze. The very sequence of data representations, X_{test}^l , is a direct product of the network's architecture and its trained parameters. Our methodology analyzes feed-forward networks, where data passes sequentially from layer l to $l + 1$ without cycles.

The transformation from one layer to the next, $T_l : x_{l-1} \mapsto x_l$, is the fundamental operation that "flows" the geometry. This transformation is composed of two distinct steps:

1. Linear Transformation: First, an affine (linear) transformation is applied using the layer's trained weight matrix W_l and bias vector b_l . This step rotates, scales, and translates the data representation.

$$h_l = W_l x_{l-1} + b_l \quad (4.19)$$

2. Non-linear Activation: Second, a non-linear activation function ψ is applied element-wise. We use the Rectified Linear Unit (ReLU), defined as:

$$x_l = \psi(h_l) = \max(h_l, 0) \quad (4.20)$$

This two-step process, $x_l = T_l(x_{l-1})$, is repeated at each layer, generating the sequence of discrete manifolds $(\mathcal{G}_k(X_{\text{test}}^1), \dots, \mathcal{G}_k(X_{\text{test}}^L))$ that we analyze using the Ricci coefficient.

4.2.1 Training objective

The network’s parameters (W_l, b_l) are learned for a binary classification task. All hidden layers use ReLU activations after their affine transformations. The output layer is a single-neuron binary classifier with a sigmoid activation, so the final output is interpreted as the probability of belonging to the positive class. The weights are optimized using the RMSprop optimizer to minimize the binary cross-entropy loss function.

4.2.2 Experimental architectures

Two related but distinct architecture descriptions are used in this project. In the global Ricci experiments, the terms narrow, wide, and bottleneck follow the architecture families used by Baptista et al. [1]. In the later local and predictive experiments, the terms flat and bottleneck describe a broader architecture grid used to test depth and width systematically.

For the global Ricci analysis, we analyzed six specific DNN architectures (investigated at 5-layer and 11-layer depths), consistent with the reference study:

- **Narrow:** Architectures with a uniform width of 25 neurons in all hidden layers.
- **Wide:** Architectures with a uniform width of 50 neurons in all hidden layers.
- **Bottleneck:** Architectures that compress information from 50-neuron layers to a central 25-neuron bottleneck

For the local Ricci and predictive experiments, we evaluate a more comprehensive set of architectures. In total, 45 distinct DNN configurations are analyzed, categorized into two main structural paradigms:

- **Flat (Uniform) Architectures:** These architectures maintain a constant width across all hidden layers. To assess varying capacity levels, we consider networks with 16, 32, 64, and 128 neurons per layer. Each width configuration is evaluated across depths ranging from 4 to 12 layers, yielding 36 distinct flat models.
- **Bottleneck Architectures:** These models follow a symmetric, autoencoder-like structure where the layer width progressively increases to a central maximum and then decreases towards the output. The maximum central width is set to 128 neurons, with the outer layers compressed down to 8 neurons. Evaluated across the same depth range (4 to 12 layers), this provides 9 additional configurations.

This structured variation separates the effect of layer width and depth from the effect of an internal bottleneck topology, allowing us to analyse how representational capacity and compression influence the geometric evolution.

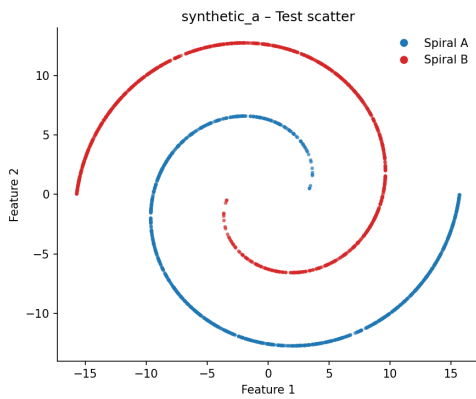
4.3 Data Sets

To ensure the robustness, a diverse collection of synthetic and real-world data sets was utilized, each presenting distinct geometric or topological challenges.

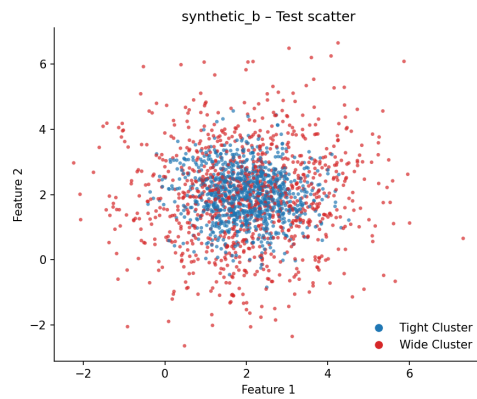
4.3.1 Synthetic data sets

Three synthetic binary classification problems were used:

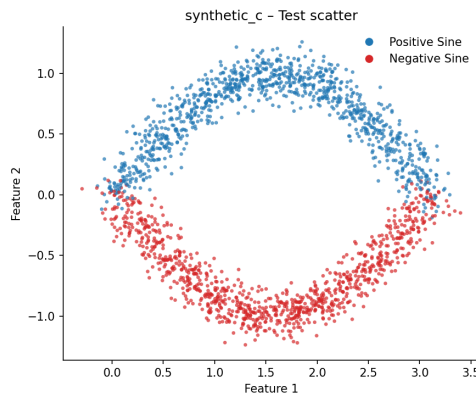
- **Data Set A:** Geometrically entangled but topologically separable spirals.
- **Data Set B:** Topologically entangled manifolds requiring a change in topology to be separated.
- **Data Set C:** Intersecting, non-separable manifolds.



(a) Data Set A: Entangled Spirals.



(b) Data Set B: Entangled Manifolds.



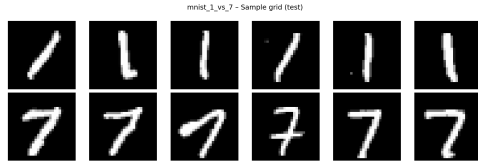
(c) Data Set C: Intersecting Manifolds.

Figure 4.1. Visualizations of synthetic data sets presenting topological challenges.

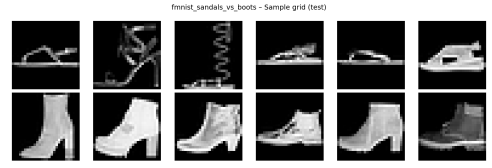
4.3.2 Real-world data sets

Two standard benchmark image data sets were used to test the method on high-dimensional, perceptually similar classes:

- **MNIST:** A dataset of handwritten digits. The binary classification tasks focused on separating '1' vs '7' and '6' vs '8'.
- **FMNIST (Fashion-MNIST):** A dataset of clothing items. The tasks focused on separating 'Sandals' vs 'Ankle Boots' and 'Shirts' vs 'Coats' and 'Sandals' vs 'Sneakers'.



(a) Sample digits from MNIST.



(b) Sample items from Fashion-MNIST.

Figure 4.2. Sample images from the real-world benchmark datasets used for binary classification tasks.

The global Ricci experiments follow the partitioning strategy of Baptista et al. [1]. For each synthetic data set, 1,000 training points and a disjoint 1,000-point test set were used. For MNIST, the standard train/test split was filtered to the selected digit pairs, and for Fashion-MNIST the standard training set contained 6,000 images per selected class while the test set contained 1,000 images per selected class. In the later local and predictive experiments, balanced binary subsets were used, with test activations extracted from held-out samples for graph construction and Ricci analysis.

5. TASKS ACCOMPLISHED

This section presents the experimental results obtained from analyzing deep neural networks through the lens of discrete Ricci flow. The experiments were designed to validate the hypothesis that DNNs exhibit Ricci flow-like geometric transformations, and to quantify the relationship between this geometric behavior and classification performance.

5.1 Global Ricci-Based Experiments

The global Ricci-based experiments focus on the relationship between total graph curvature, total geodesic distance change, and classification accuracy across datasets and architectures.

5.1.1 Experimental configuration

Table 5.1 summarizes the experimental setup used throughout this study. A total of 214 configurations were tested, each involving the training of 70 independent models to ensure statistical robustness. The experiments covered five distinct architectures across two depth levels (5 and 11 layers), with varying k -NN neighborhood sizes for both synthetic and real-world datasets.

Table 5.1. Summary of All Experimental Configurations

Parameter	Value
Number of Models per Configuration (b)	70
Training Epochs	50
Batch Size	32
Accuracy Threshold	98%
Optimizer	RMSprop
Activation Function	ReLU
Loss Function	Binary Cross-Entropy
Tested Architectures	Narrow (5, 11), Wide (5, 11), Bottleneck (5)
k-NN Values (Synthetic)	6, 7, 9, 10, 15, 18, 20, 30, 50, 90, 100
k-NN Values (Real)	325, 400, 500
Total Configurations	214

5.1.2 Depth-based analysis of Ricci flow behavior

A central finding of this study is that deeper networks exhibit stronger Ricci flow-like behavior. Table 5.2 presents the correlation analysis between geometric flow characteristics and classification accuracy for networks of different depths. In this table, N denotes the number of experimental configurations (i.e., unique combinations of dataset, architecture, and k -NN value) analyzed for each depth category.

Table 5.2. Depth-Based Correlation Analysis on Real Datasets (MNIST - FMNIST)

Depth	N	Accuracy (mean $\pm$ std)	Pearson r	k
5	24	98.21% $\pm$ 0.29%	-0.59	500
11	24	98.88% $\pm$ 0.26%	-0.722	500

The results confirm the theoretical expectation: 11-layer networks demonstrate stronger geometric flow behavior compared to 5-layer networks, as evidenced by the higher mean accuracy (98.88% vs. 98.21%). Furthermore, the correlation between geometric flow characteristics and accuracy is statistically significant for both depths, with deeper networks showing stronger correlation (Pearson $r = -0.722$, $p < 0.001$) compared to shallower networks (Pearson $r = -0.59$, $p = 0.002$). This finding aligns with the framework of Baptista et al. [1], suggesting that the additional layers provide more opportunities for the network to “smooth” the data geometry through Ricci flow-like transformations.

5.1.3 Architecture comparison

Table 5.3 presents a detailed comparison of the five tested architectures in terms of their classification performance across different depth and width configurations.

Table 5.3. Architecture-Based Summary of Classification Accuracy (MNIST)

Architecture	Depth	Width	Accuracy (%)
Narrow	5	25	98.37 $\pm$ 0.28
Narrow	11	25	98.69 $\pm$ 0.30
Wide	5	50	98.70 $\pm$ 0.26
Wide	11	50	98.99 $\pm$ 0.24
Bottleneck	5	50 $\rightarrow$ 25	98.61 $\pm$ 0.27

Several patterns emerge from this analysis. The Wide 11-layer architecture achieves the highest accuracy (98.99%), demonstrating the synergistic effect of depth and capacity. Increasing depth consistently improves accuracy: Narrow networks improve from 98.37% to 98.69%, and Wide networks from 98.70% to 98.99%. The Bottleneck architecture achieves accuracy (98.61%) despite having only 5 layers, suggesting that information compression contributes to effective geometric simplification. Finally, wider networks consistently outperform narrower ones at the same depth.

5.1.4 Dataset-specific results

Table 5.4 identifies the best-performing configuration for each dataset-architecture combination, providing insights into how task difficulty influences both classification performance and geometric flow behavior.

Table 5.4. Results for Each Dataset-Architecture Combination

Dataset	Architecture	k -Value	Ricci ρ	Accuracy (%)
<i>MNIST</i>				
1 vs 7	Wide 11	500	−0.779	99.07
1 vs 7	Narrow 11	500	−0.697	99.04
1 vs 7	Narrow 5	500	−0.640	98.94
6 vs 8	Wide 11	500	−0.797	98.94
6 vs 8	Narrow 11	500	−0.671	98.84
6 vs 8	Narrow 5	500	−0.625	98.81
<i>Fashion-MNIST</i>				
Sandals vs Boots	Bottleneck 5	500	−0.736	98.41
Sandals vs Boots	Wide 5	500	−0.654	98.48
Sandals vs Boots	Wide 11	500	−0.635	98.50
Shirts vs Coats	Narrow 11	500	−0.238	98.42
Shirts vs Coats	Bottleneck 5	500	−0.208	98.32
Shirts vs Coats	Wide 11	500	−0.192	98.53

The results reveal a clear relationship between task complexity, Ricci flow strength, and classification performance. For the MNIST tasks, strong negative Ricci coefficients (ρ ranging from −0.625 to −0.797) accompany high accuracy values up to 99.07%. In contrast, the Shirts vs Coats task shows notably weaker Ricci coefficients (approximately −0.2) despite achieving comparable accuracy, suggesting that perceptually similar classes may require different representational strategies. The consistent selection of $k = 500$ as the optimal neighborhood size indicates that larger-scale geometric structures are more informative for high-dimensional image data.

5.2 Local Ricci-Based Analysis

In addition to the global Ricci coefficient used in the main analysis, we examine curvature behavior at a more local level by considering layer-wise and training-based variations of discrete Ricci curvature. This perspective follows the local graph-based framework discussed in [34], where curvature is evaluated directly on neighborhood structures and across individual layer transitions. The aim is to observe how geometric changes develop within the network beyond a single global summary.

5.2.1 Experimental configuration

The local Ricci-based analysis is conducted under configurations designed to examine curvature behavior both across layers and during training. For the layer-wise analysis, a fixed-width fully connected architecture with 25 neurons per layer is used, and networks of varying depth $L \in \{10, 12, 14, 16, 18\}$ are evaluated. Each configuration is trained for 50 iterations, and curvature values are computed using an approximate Ollivier-Ricci formulation.

Table 5.5. Summary of Local Ricci Experimental Configurations

Parameter	Value
Architecture (Layer-wise)	Fully connected
Width	25 neurons per layer
Depths	10, 12, 14, 16, 18
Iterations	50
Curvature Type	Approximate Ollivier-Ricci
Architectures (Training-based)	45 (flat + bottleneck)
Depth Range	4–12
Widths	16, 32, 64, 128
Training Mode	Epoch-based
Comparison	Standard early stopping vs curvature behavior

In addition, training-based experiments are carried out to analyze the relationship between curvature evolution and convergence. These experiments are conducted on the set of 45 architectures described in Section 4.2.2, including both flat and bottleneck configurations. In this case, curvature values are tracked across training epochs and compared with standard early stopping criteria.

5.2.2 Layer-wise curvature behavior

To analyze how curvature evolves across layers, we evaluate the layer-wise Ricci coefficient for networks of varying depth using the configuration described above. The experiment follows a binary classification task (sandal vs sneaker), with the setup used in [34].

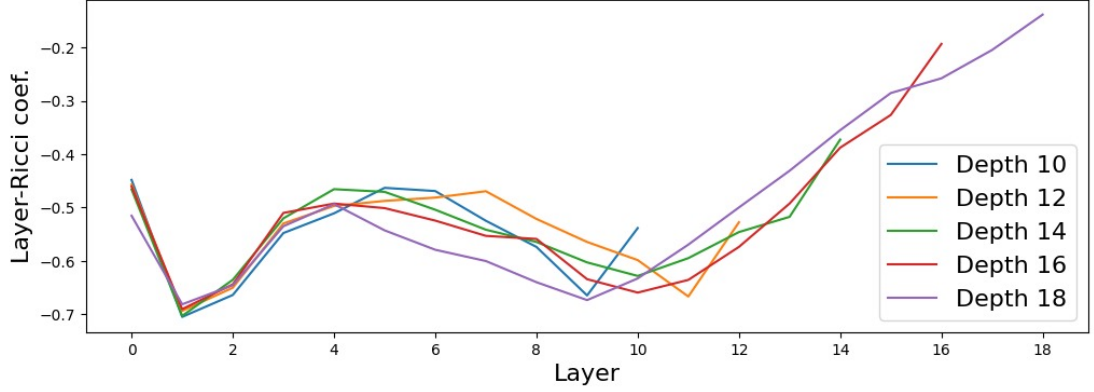


Figure 5.1. Layer-Ricci coefficients evaluated across successive layers for flat architectures (25 neurons per layer) at varying depths (10, 12, 14, 16, 18). The values are derived using the approximate Ollivier-Ricci curvature after 50 training iterations.

Figure 5.1 shows the evolution of the layer Ricci coefficient as a function of layer index for each depth configuration. It can be observed that deeper networks exhibit a more gradual and structured transition in curvature values across layers, while shallower networks display sharper variations. In particular, curvature values tend to decrease in early layers and then stabilize or increase in later layers, indicating a progressive reorganization of the feature space.

This behavior is consistent with the interpretation that deeper architectures allow a more continuous geometric transformation of the data manifold. Compared to the global Ricci coefficient, the layer-wise analysis provides a more detailed view of how geometric changes are distributed within the network.

5.2.3 Early stopping analysis

To examine the relationship between geometric evolution and training dynamics, we analyze curvature behavior across training epochs and compare it with standard early stopping criteria. The experiments are conducted on the set of 45 architectures described in Section 4.2.2.

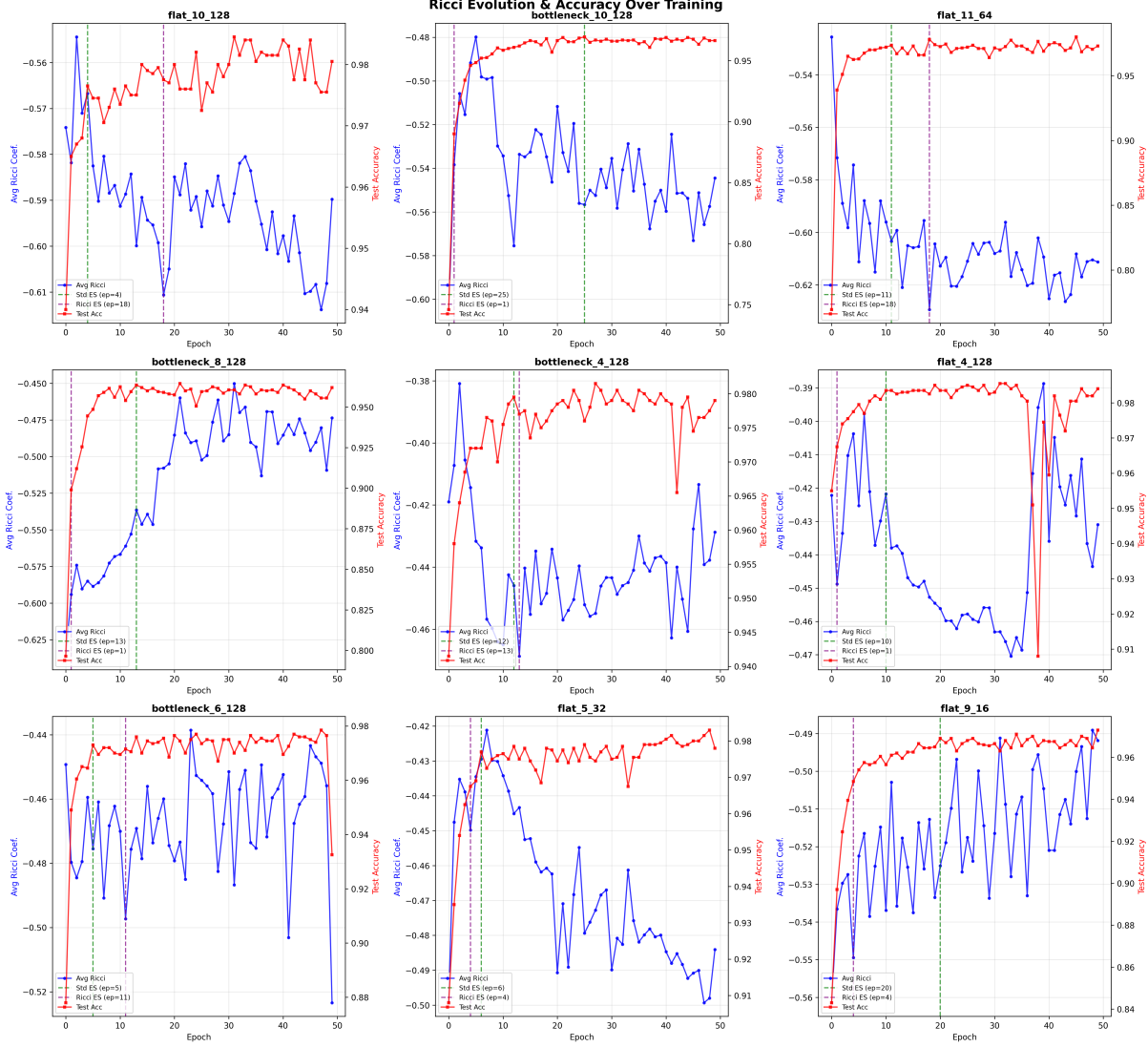


Figure 5.2. Evolution of the average local Ricci coefficient computed with Augmented Forman-Ricci curvature (blue) and test accuracy (red) across training epochs for various flat and bottleneck architectures. Green dashed lines indicate standard early stopping (Std ES) epochs, while purple dashed lines indicate Ricci-based early stopping (Ricci ES).

Figure 5.2 shows representative examples of curvature evolution and test accuracy over training epochs for different architectures. It can be observed that curvature values exhibit significant oscillation throughout training, particularly in the early epochs, and do not consistently converge to a stable value within the observed training window. This volatility makes it non-trivial to define a reliable stopping criterion directly from the raw curvature signal.

To address this, both the standard and Ricci-based early stopping criteria are configured to ignore the first observed qualifying change, activating only upon a subsequent, more sustained signal. Under this setting, the Ricci-based criterion still tends to trigger earlier than standard early stopping across most architectures, suggesting that the geometric signal captures a meaningful shift in training dynamics that precedes the plateau observable in validation performance. In bottleneck architectures, this difference is more pronounced, likely due to the stronger curvature gradients induced by the compression layers.

These findings demonstrate that raw local curvature measurements should not be positioned as a standalone replacement for standard early stopping. Instead, the local Ricci coefficient is best utilized as a highly valuable, complementary diagnostic tool. It provides critical theoretical insights into the exact moments a network transitions from geometric manifold smoothing into structural overfitting and topological collapse.

5.2.4 Averaging strategies for layer-wise Ricci coefficients

This experiment investigated whether robust averaging methods could improve the stability of Ricci-based early stopping. As discussed in Figure 4, the layer-wise Ricci coefficients tend to become worse in later layers, especially for deeper networks. In this context, “worse” refers to Ricci coefficients moving toward less negative values. Since more negative Ricci coefficients are interpreted as indicating stronger geometric organization, this late-layer increase can distort the average Ricci signal.

The original Ricci-based early stopping criterion used a single average Ricci coefficient over all layer transitions. However, a plain arithmetic mean treats all layers equally:

$$\bar{r} = \frac{1}{L} \sum_{l=1}^L r_l,$$

where r_l is the Ricci coefficient of layer transition l and L is the number of layer transitions. This becomes problematic when the final layers produce unstable or degenerate Ricci values. These layers are close to the output space and may reflect compression into the classifier rather than the geometry of the learned representation.

To reduce the influence of these unstable layers, several robust averaging methods were tested.

The first method used IQR-based averaging. It used the interquartile range to exclude high-side outliers. Since problematic Ricci values are those closer to zero, only unusually high Ricci coefficients were removed:

$$r_l > Q_3 + 1.5 \cdot IQR.$$

The average was then computed only over the retained layers.

The second method used MAD-based averaging. It used the median absolute deviation. A layer was excluded if its Ricci coefficient was sufficiently far above the median:

$$r_l > \text{median}(r) + z \cdot 1.4826 \cdot MAD.$$

This method is less sensitive to extreme values.

The progressive trimming method iteratively removed the highest Ricci value, that is, the layer closest to zero, until the remaining coefficients formed a more stable cluster or a maximum trimming ratio was reached.

The tail-drop method removed the final one or two layer transitions depending on network depth. This was based on the observation that output-adjacent layers often produced unreliable Ricci coefficients.

The final method used threshold-based averaging with a fixed threshold. Only layers satisfying

$$r_l < -0.6$$

were included in the average. Layers with Ricci coefficients greater than or equal to this threshold were treated as geometrically unreliable.

The experimental output was obtained after applying the robust Ricci averaging procedure. The early stopping results were compared against standard patience-based early stopping. The experiment was conducted over 45 architectures.

Metric	Result
Total architectures	45
Ricci-based early stopping earlier	30
Standard early stopping earlier	15
Ties	0
Average accuracy with standard early stopping	0.9734
Average accuracy with Ricci-based early stopping	0.9479

Table 5.6. Summary of early stopping results after robust Ricci averaging.

The results show that Ricci-based early stopping triggered earlier in most architectures. However, this earlier stopping also produced a lower average accuracy than standard early stopping. This suggests that the robust Ricci average made the stopping signal more sensitive, but not necessarily more accurate as a final training criterion.

Although these robust averaging methods reduced the influence of unstable late-layer Ricci coefficients, they were not used in the later experiments. The main limitation is that these methods treat layer-wise Ricci coefficients as an unordered collection of values. In a neural network, however, layer order is essential. Training proceeds from the input layer through intermediate representations and finally to the output layer. Therefore, the position of a Ricci coefficient is part of its meaning.

For example, IQR-based or MAD-based filtering may remove a Ricci coefficient because it is an outlier, but it does not consider whether that coefficient comes from an early, middle, or late layer. Similarly, progressive trimming removes the largest Ricci values based only on magnitude. This can hide the fact that the deterioration happens systematically near the end of the network. In other words, the late-layer worsening is not merely noise; it is an ordered depth-dependent pattern.

Therefore, while robust averaging helped produce a cleaner scalar Ricci signal, it also removed information about how the geometry evolves across layers. Since the later experiments focused on understanding Ricci curvature as a depth-dependent geometric measure, preserving layer order was more important than obtaining a single filtered average. For this reason, these methods were treated as exploratory experiments and were not adopted in the final experimental pipeline.

5.2.5 Two-phase Ricci-based architecture screening: a potential early indicator

As observed in the preliminary early stopping analysis, curvature values often reach a stopping condition earlier than standard validation-loss heuristics. This early stopping behavior raises a theoretical question: might the geometric stabilization signal, captured by the Ricci coefficient, serve as a potential early indicator of the network’s ultimate performance capability?

To investigate this hypothesis, a two-phase architecture screening experiment was conducted using the Fashion-MNIST dataset (Sneakers vs. Sandals) across a grid of 45 flat and bottleneck configurations. The augmented Forman-Ricci curvature was utilized with a neighborhood size k set to 5% of the test set size.

The experimental protocol was divided into two distinct phases:

- **Phase 1 (Probe Run):** All 45 architectural configurations were trained for a fixed duration of exactly 10 epochs. At the end of this phase, the robust average layer Ricci coefficient was computed. A selection threshold of -0.6 was established. Architectures yielding an average Ricci score greater than this threshold (closer to 0) were flagged as geometrically immature, suggesting they had not yet initiated curvature-driven flow, and were selected for the next phase.
- **Phase 2 (Extended Run):** The 8 architectures flagged in Phase 1 were completely re-initialized with fresh weights to ensure a fair comparison, and trained for an extended budget of 200 epochs. During this phase, both standard validation accuracy-based early stopping (Std ES) and Ricci-based early stopping (Ricci ES) were deployed as passive observers. They recorded the epochs at which each criterion would have selected a stopping point and the corresponding test accuracies, without interfering with the 200-epoch run.

Figure 5.3 illustrates the trajectory of both the validation accuracy and the average Ricci coefficient over the extended 200-epoch training for representative architectures. By evaluating the final performance at epoch 200 against the theoretical stopping points flagged by the passive observers, we can analyze whether the early accuracy achieved at the Ricci ES trigger aligns with the network’s peak performance.

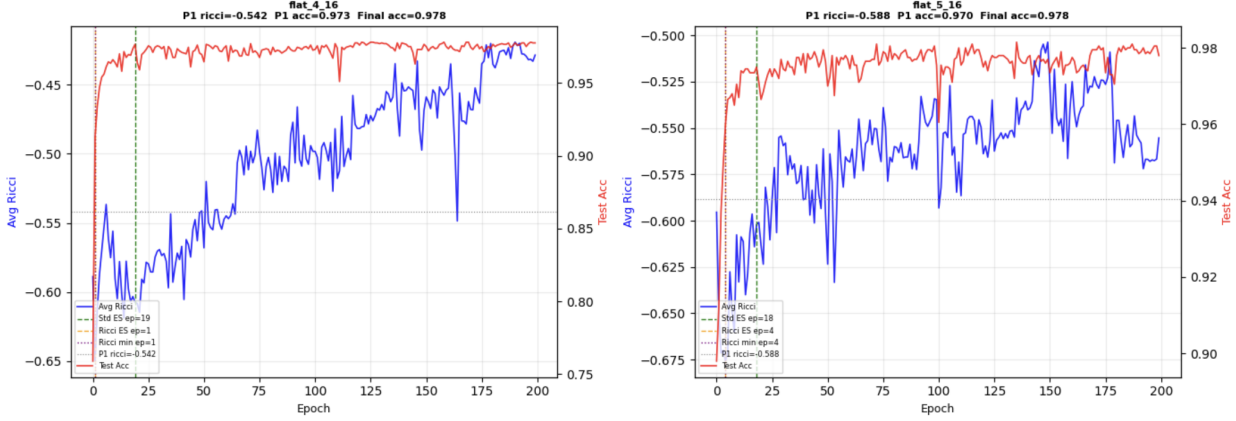


Figure 5.3. Evolution of the average local Ricci coefficient and test accuracy over 200 epochs for representative architectures. The graph highlights the theoretical stopping points for Ricci-based early stopping and standard early stopping during the extended run.

The quantitative results of this extended tracking are summarized in Table 5.7.

Table 5.7. Summary of Two-Phase Architecture Screening Results

Architecture	Phase 1 Ricci	Phase 1 Acc. (%)	Std ES Epoch	Std ES Acc. (%)	Ricci ES Epoch	Ricci ES Acc. (%)	Final Acc. (200 Ep) (%)
flat_4_16	-0.5419	97.30	19	97.65	1	91.30	97.75
flat_5_16	-0.5882	96.95	18	97.50	4	96.00	97.80
flat_7_16	-0.5750	95.35	8	96.15	1	86.60	96.85
flat_10_64	-0.5888	96.75	25	97.50	4	96.30	97.65
flat_12_128	-0.5881	96.90	6	96.90	15	97.35	98.30
bottleneck_7_128	-0.5882	97.10	11	96.90	1	94.25	97.60
bottleneck_10_128	-0.5930	96.10	23	97.25	7	95.10	97.25
bottleneck_12_128	-0.5963	95.60	25	95.90	2	86.05	96.70
Mean Values	-0.5809	96.51	16.8	96.97	4.3	92.87	97.49

The empirical data provides an interesting perspective on the potential predictive nature of discrete Ricci flow. As shown in Table 5.7, the Ricci ES mechanism triggered at an early average of 4.3 epochs, preceding the standard early stopping metric which required an average of 16.8 epochs.

Despite this early signal, the networks had achieved a mean accuracy of 92.87% at the Ricci ES epoch. When compared to the final 200-epoch mean accuracy of 97.49%, it appears that Ricci ES might capture a substantial portion of the network’s learning potential relatively early in the training process. For certain higher-capacity architectures, such as flat_12_128, the similarity is notable, with Ricci ES triggering at epoch 15 (97.35% accuracy) compared to the final 200-epoch accuracy (98.30%).

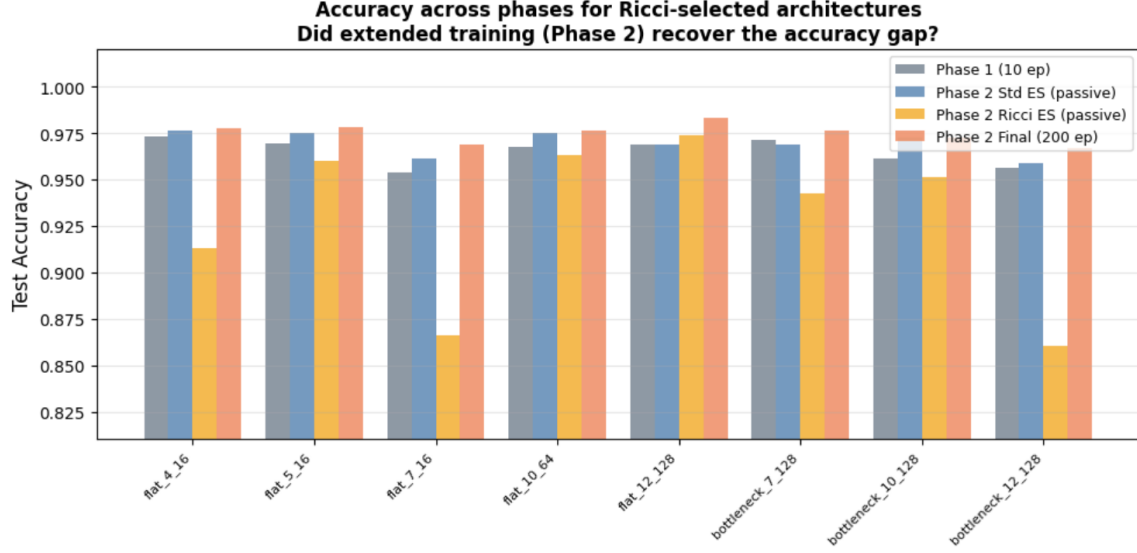


Figure 5.4. Comparison of test accuracies across training phases for the geometrically immature architectures. The chart illustrates whether the extended 200-epoch training recovered the performance gap observed at the early stopping triggers.

However, it is important to observe that for highly compressed models like bottleneck 12_128, the Ricci ES triggered much earlier, resulting in a larger performance gap (86.05% vs. 96.70%). This suggests that while geometric manifold smoothing shows promise as an early indicator of representational capability, its current form might be too aggressive for certain architectural constraints, particularly those with severe topological bottlenecks.

In conclusion, while the early signals driven by Ricci curvature appear promising in reducing computational overhead and estimating eventual success, the results indicate that they should be interpreted with caution. The two-phase screening demonstrates its potential as an effective architecture filtering tool, yet its aggressiveness for highly compressed models highlights the need for further architectural calibration. However, this capacity to signal long-term potential naturally leads to a more ambitious question: If early curvature dynamics can identify whether an architecture will eventually stabilize, can they also predict how well it will perform relative to others? To explore whether a short initial geometric probe can reliably rank a diverse set of architectures by their eventual accuracy, the following section extends this predictive analysis across the full suite of models without early stopping interventions.

5.2.6 Predictive Ricci Analysis

This experiment investigates whether the Ricci coefficient measured during the early stages of training carries predictive information about an architecture’s long-term classification performance. While the preceding experiments analyzed curvature behavior across layers or in relation to early stopping dynamics, the focus here shifts to a different question: can a short initial geometric probe, conducted over the first few epochs, reliably rank architectures by their eventual accuracy after extended training? In practical terms, the objective is to determine whether these early geometric signals can identify architectures that are worth allocating additional training resources to.

Fashion-MNIST

To answer this, all 45 architectures (36 flat and 9 bottleneck) were trained for a full 200 epochs on the Fashion-MNIST binary classification task (Sneakers vs. Sandals) without restarts or early stopping.

The experimental configuration used the Adam optimizer with a learning rate of 0.001 and a batch size of 512. Curvature was computed using Augmented Forman-Ricci, and the k -NN neighborhood parameter was set to $k = 100$ (5% of the 2 000 test samples).

The training process was divided into two phases:

- **Ricci probe window (epochs 0–9):** At every epoch, hidden-layer activations were extracted from the test set, k -NN graphs were constructed, all-pairs shortest-path distances were computed, and the Augmented Forman-Ricci curvature was calculated. From these, the layer Ricci coefficient was obtained for each layer transition and the mean across all layers was recorded.
- **Accuracy-only phase (epochs 10–199):** For the remaining 190 epochs, only training and test accuracy were tracked. No curvature computations were performed, reducing runtime significantly while preserving the full performance trajectory.

Table 5.8. Experimental parameters for the 200-epoch Fashion-MNIST predictive Ricci experiment.

Parameter	Value
Dataset	Fashion-MNIST (Sneakers vs. Sandals)
Number of architectures	45 (36 flat + 9 bottleneck)
Total training epochs	200
Ricci computation window	Epochs 0–9
Optimizer	Adam
Learning rate	0.001
Batch size	512
Curvature type	Augmented Forman-Ricci
k -NN parameter	$k = 100$ (5% of test set)

For each architecture, two scalar summary metrics were computed after the full training run:

1. **Early Ricci score:** The arithmetic mean of the average layer Ricci coefficient across the first 10 epochs (epochs 0–9). A more negative value indicates stronger Ricci flow-like behavior during the initial training phase.
2. **Long-term accuracy:** The arithmetic mean of the test accuracy across all 200 epochs. Unlike the final-epoch accuracy, this cumulative measure rewards architectures that learn quickly and maintain stable performance throughout training.

The results show that across the 45 architectures, long-term accuracy values ranged from approximately 0.92 to 0.98, while early Ricci scores ranged from approximately -0.71 to -0.40 . Flat architectures with moderate depth (6–9 layers) and sufficient width (64–128 neurons) generally achieved the highest long-term accuracy. Among bottleneck architectures, shallower configurations (4–6 layers) performed comparably to flat architectures, whereas deeper bottleneck configurations (10–12 layers) exhibited lower long-term accuracy, likely due to information compression at greater depths.

The central result is the statistically significant negative correlation between early rank based Ricci scores and long-term accuracy, summarized in Table 5.9.

Table 5.9. Correlation between early Ricci score and long-term accuracy ($n = 45$ architectures).

Metric	Value	p -value
Pearson r	+0.505	0.0004
Spearman ρ	+0.572	< 0.0001
Kendall τ	+0.392	0.0001

All three measures are positive and statistically significant ($p < 0.001$). The direction of the correlation is consistent with the hypothesis: a more negative early Ricci score, indicating stronger geometric flow during the first 10 epochs, tends to co-occur with higher long-term accuracy. However, the magnitudes are moderate. The Pearson correlation ($r = +0.505$) indicates that the early Ricci score accounts for roughly 25% of the variance in long-term accuracy ($r^2 \approx 0.255$), leaving the majority of performance variation unexplained. The Spearman rank correlation ($\rho = +0.572$) is somewhat stronger, suggesting that the relationship may be better described as a monotonic trend than a linear one, though still far from a strong predictor. The Kendall-tau coefficient ($\tau = +0.392$) is the weakest of the three, indicating limited concordance between the two rankings.

Figure 5.5 visualizes the relationship between early Ricci score and long-term accuracy for all 45 architectures.

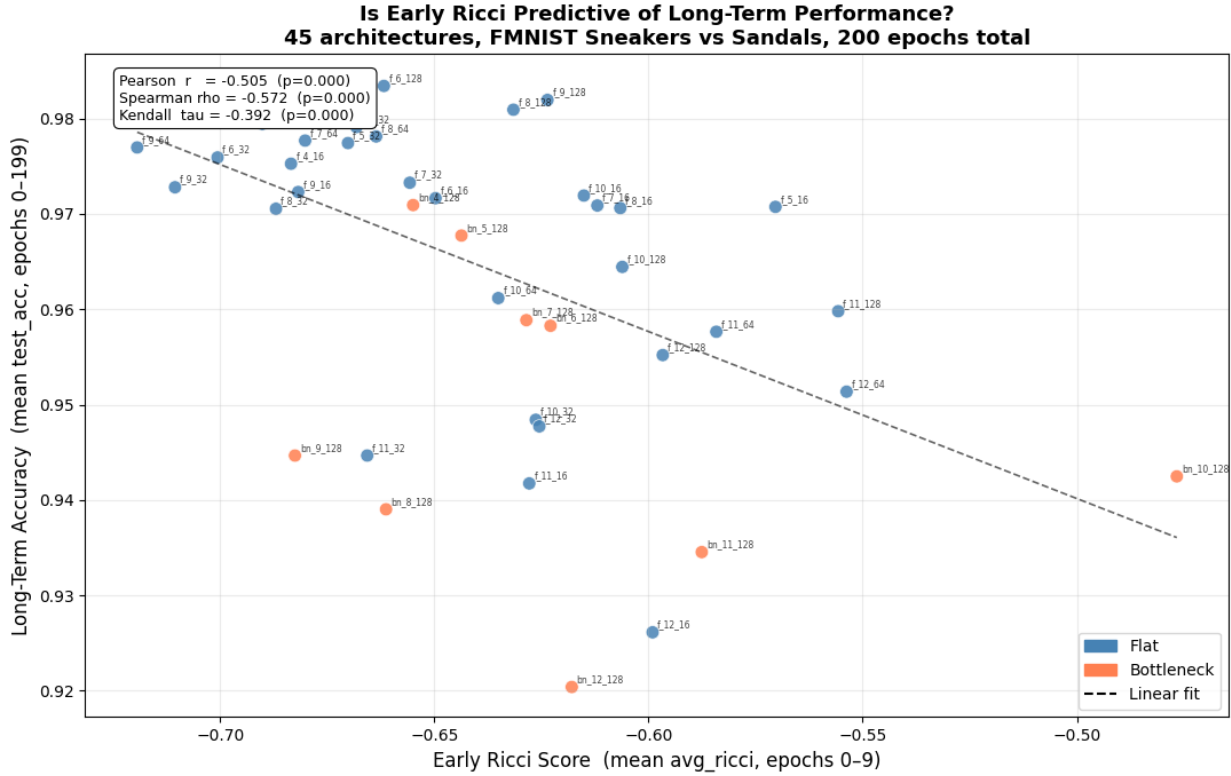


Figure 5.5. Scatter plot of early Ricci score versus long-term accuracy for 45 architectures trained on Fashion-MNIST (Sneakers vs. Sandals) for 200 epochs. Blue points represent flat architectures, orange points represent bottleneck architectures. The dashed line shows the linear fit. Correlation coefficients are shown in the upper-left corner.

Table 5.10 lists the ten highest-ranked architectures by long-term accuracy and their corresponding early Ricci scores.

Table 5.10. Top 10 architectures ranked by long-term accuracy.

LT Acc Rank	Architecture	Early Ricci	Ricci Rank	Long-term Acc
1	flat_4_128	-0.678	11	0.984
2	flat_6_128	-0.662	19	0.983
3	flat_5_128	-0.693	4	0.983
4	flat_7_128	-0.675	13	0.982
5	flat_9_128	-0.624	31	0.982
6	flat_8_128	-0.632	26	0.981
7	flat_5_64	-0.672	14	0.980
8	flat_6_64	-0.676	12	0.980
9	flat_4_64	-0.690	5	0.979
10	flat_4_32	-0.668	16	0.979

The architecture with the strongest early Ricci score (flat_9_64, early Ricci = -0.719 , Ricci rank 1) achieved a long-term accuracy of 0.977 (LT Acc rank 14). While this places it among the top performers, the overall rank agreement is limited. For instance, flat_9_128 ranks 31st by Ricci score yet achieves a high long-term accuracy of 0.982 (LT Acc rank 5). Conversely, bottleneck architectures such as bn_9_128 have a strong early Ricci score (rank 8) but rank only 39th in long-term accuracy. These rank mismatches indicate that the early Ricci score alone is not a reliable predictor of final performance for individual architectures.

Among the bottleneck architectures, deeper configurations (depth 10–12) consistently produce lower long-term accuracy values (0.920–0.943) and their early Ricci scores cluster in the range -0.48 to -0.62 , indicating weaker early geometric organization. This pattern suggests that information compression at greater depths may impede both early geometric reorganization and long-term learning capacity, though the sample of bottleneck architectures ($n = 9$) is too small to draw strong conclusions.

The observed correlations are statistically significant but moderate in magnitude. The early Ricci score explains only about a quarter of the variance in long-term accuracy, indicating that the geometric signal captured during the first 10 epochs is at best a partial indicator of eventual performance. Factors such as optimizer dynamics, batch normalization effects, and architecture-specific convergence behavior likely account for the remaining variance.

Additionally, the experiment was conducted on a single binary classification task (Fashion-MNIST, Sneakers vs. Sandals), and it is unclear whether the observed correlation would hold on other datasets or more complex classification problems. The choice of 10 epochs as the probe window was motivated by computational constraints, and different window lengths could yield different correlation strengths.

The rank-level analysis further highlights the limitation: while the overall trend is consistent with the hypothesis, individual architecture predictions based on early Ricci scores are unreliable. High-ranking architectures by early Ricci do not necessarily correspond to high-ranking architectures by accuracy, as illustrated by several notable mismatches in Table 5.10.

In summary, the results suggest a weak-to-moderate association between early geometric behavior and long-term accuracy. The early Ricci score may provide a coarse signal for filtering candidate architectures, but it should not be used as a standalone criterion for architecture selection. Verification on additional datasets is necessary to assess the robustness of this finding.

CIFAR-10

Following the methodology of the Fashion-MNIST predictive experiment, this analysis shifts the focus to the CIFAR-10 dataset (specifically, the binary classification of Cars vs. Planes). The objective is to evaluate whether the predictive capacity of the local Ricci coefficient holds on a more complex, natural image task. Recognizing that a 10-epoch probe might be insufficient for capturing the onset of geometric stabilization in a more challenging dataset, the observation window was extended to the first 40 epochs.

The experimental configuration was similar to the previous setup. All 45 architectures (36 flat and 9 bottleneck) were trained for a full 200 epochs without restarts or early stopping. The Adam optimizer was used with a learning rate of 0.001 and a batch size of 512. Curvature was computed using the Augmented Forman-Ricci formulation, and the k -NN neighborhood parameter was initially adjusted to $k = 50$, corresponding to 5% of the 1 000 balanced test samples.

The training process was divided into two phases:

- **Extended Ricci probe window (epochs 0–39):** At every epoch, hidden-layer activations were extracted, k -NN graphs were constructed, and the Augmented Forman-Ricci curvature was calculated. From these, the layer Ricci coefficient was obtained for each layer transition and the mean across all layers was recorded.
- **Accuracy-only phase (epochs 40–199):** For the remaining 160 epochs, only the validation and training accuracies were tracked to minimize computational overhead while capturing the full performance trajectory.

As in the previous experiment, two scalar summary metrics were derived for each architecture after the full training run:

1. **Early Ricci score:** The arithmetic mean of the average layer Ricci coefficient across the first 40 epochs.
2. **Long-term accuracy:** The arithmetic mean of the test accuracy across all 200 epochs.

The quantitative relationship between the early Ricci scores (ranked such that a more negative score is considered geometrically stronger) and the long-term accuracy is detailed in Table 5.11.

Table 5.11. Correlation between early Ricci score and long-term accuracy on CIFAR-10 ($n = 45$ architectures).

Metric	Value	p -value
Pearson r	+0.2305	0.1276
Spearman ρ	+0.2572	0.0881
Kendall τ	+0.1636	0.1130

To visually assess this relationship, Figure 5.6 plots the early Ricci scores against the long-term accuracy for all evaluated architectures.

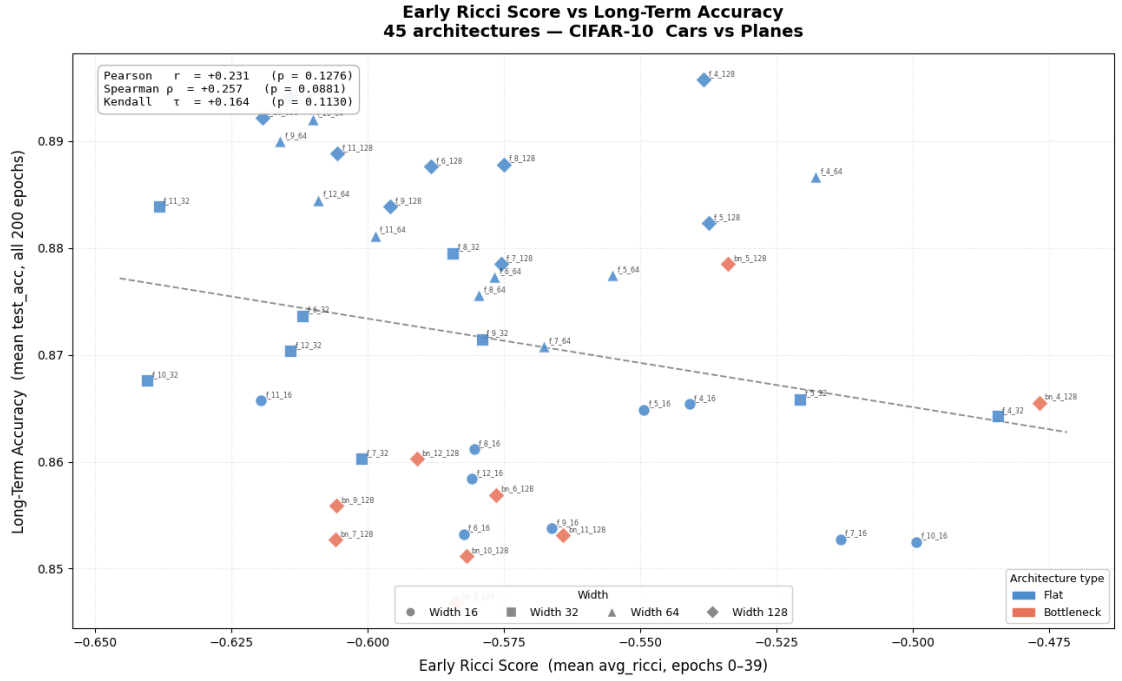


Figure 5.6. Scatter plot of early local Ricci score versus long-term accuracy for 45 architectures trained on CIFAR-10 (Cars vs. Planes). The plot illustrates the weak and noisy correlation observed in natural image classification tasks.

Furthermore, to ensure that these weak correlations are not merely an artifact of the specifically chosen neighborhood size ($k = 50$), Table 5.12 presents a parameter sweep across various k -NN values.

Table 5.12. Rank correlation of Local Ricci with long-term accuracy across various k -NN values on CIFAR-10.

k -Value	n	Pearson r	Spearman ρ	Kendall τ
$k = 5.0\%$	45	+0.2572	+0.2572	+0.1636
$k = 3.0\%$	45	+0.2111	+0.2111	+0.1354
$k = 7.0\%$	45	+0.2462	+0.2462	+0.1576

In contrast to the Fashion-MNIST findings, the correlations observed on the CIFAR-10 dataset are weak and lack statistical significance. While the positive direction of the coefficients (ranging from +0.16 to +0.25) indicates a marginal trend where a stronger geometric signal loosely corresponds to higher eventual accuracy, the high p -values confirm that there is no reliable predictive relationship.

These results suggest that as dataset complexity increases, moving from grayscale topological boundaries (Fashion-MNIST) to natural RGB images (CIFAR-10), the early local geometric evolution becomes significantly noisier. Despite extending the observation window to 40 epochs and rigorously sweeping across multiple neighborhood scales, the early local Ricci score fails to consistently forecast long-term success. This implies that for complex visual classification tasks, early-stage local manifold smoothing is likely overridden by other training dynamics, rendering it insufficient to act as a standalone predictive oracle for architecture screening.

Global vs. Local Ricci

Building upon the local Ricci analysis, which demonstrated a noisy and weak predictive signal on the CIFAR-10 dataset, this section shifts the focus to the global Ricci flow perspective. The primary motivation is to determine whether looking at the network’s geometry as a whole, rather than focusing on local neighborhoods, can provide a better predictive signal.

When comparing different architectures, using the raw global Ricci coefficient (ρ_k) can be misleading because it is heavily affected by the network’s depth (L). Since the global Ricci score is a correlation calculated across layer transitions, a deep network (e.g., 12 layers) naturally provides more data points than a shallow network (e.g., 4 layers), creating an unfair structural bias. To solve this problem and put all architectures on a level playing field, we apply Fisher’s z -transformation [1]:

$$z_k = \frac{\operatorname{arctanh}(\rho_k)}{\sqrt{L-4}} \quad (5.1)$$

This straightforward adjustment mathematically removes the depth bias, allowing us to accurately and fairly compare the geometric flow of architectures with varying depths. However, due to the $\sqrt{L-4}$ term in the denominator, this transformation becomes mathematically undefined for architectures with 4 or fewer layers. Consequently, the shallowest networks in our grid ($L = 4$) had to be excluded from the normalized analysis, slightly reducing the evaluated sample size for the Fisher z metric.

This phase of the analysis did not require retraining the models. Instead, it reused the comprehensive set of hidden-layer activations saved during the first 40 epochs of the preceding CIFAR-10 experiment. For the eligible architectures, the global Ricci coefficient (ρ_k) and its Fisher z-transformed counterpart (z_k) were computed.

The analysis compares three distinct early geometric signals against long-term accuracy:

1. **Local Ricci Score:** The aggregated mean of local neighborhood curvatures.
2. **Global Ricci Score (ρ):** The raw depth-wise Pearson correlation.
3. **Global Ricci Score (z):** The Fisher z-transformed, depth-normalized correlation.

The results and comparison are summarized in Table 5.13, which reports the rank correlations (Pearson, Spearman, and Kendall) for the three different geometric metrics against the final 200-epoch accuracy, extracting the optimal training k baseline values.

Table 5.13. Comparison of early geometric scores (Local, Global ρ , Global z) against long-term accuracy on CIFAR-10.

Metric	Local Ricci	Global Ricci (ρ)	Global Ricci (z)
Pearson r	+0.2572	−0.3829	−0.4867
Spearman ρ	+0.2572	−0.3829	−0.4867
Kendall τ	+0.1636	−0.2566	−0.3205

Because of the ranking methodology, a positive correlation indicates that a more negative Ricci score predicts higher accuracy. Therefore, the negative values for Global Ricci metrics indicate an inverse relationship.

To assess how these predictive capacities behave across varying observation scales, Table 5.14 presents a comprehensive parameter sweep. The chart plots the correlation strength of all three metrics across different k -NN neighborhood sizes.

Table 5.14. Rank correlation of Local vs Global Ricci with long-term accuracy across various k-NN values on CIFAR-10.

Type	k-Value	n	Pearson r	p-value	Spearman ρ	p-value	Kendall τ	p-value
Local	$k = 5.0\%$	45	+0.2572	0.0881	+0.2572	0.0881	+0.1636	0.1130
Local	$k = 3.0\%$	45	+0.2111	0.1640	+0.2111	0.1640	+0.1354	0.1899
Local	$k = 7.0\%$	45	+0.2462	0.1030	+0.2462	0.1030	+0.1576	0.1270
Global r	$k = 12.5\%$	45	−0.4406	0.0025	−0.4406	0.0025	−0.3071	0.0029
Global z	$k = 12.5\%$	40	−0.5390	0.0003	−0.5390	0.0003	−0.3564	0.0012
Global r	$k = 18.5\%$	45	−0.3829	0.0094	−0.3829	0.0094	−0.2566	0.0130
Global z	$k = 18.5\%$	40	−0.4867	0.0015	−0.4867	0.0015	−0.3205	0.0036
Global r	$k = 25.0\%$	45	−0.3096	0.0385	−0.3096	0.0385	−0.1879	0.0688
Global z	$k = 25.0\%$	40	−0.4426	0.0042	−0.4426	0.0042	−0.2897	0.0085

Positive correlation indicates that a more negative early Ricci score predicts higher accuracy.

The comparative analysis presents the observed outcomes of local and global geometric flow metrics. The local Ricci approach yielded a weak, statistically insignificant positive trend ($p > 0.05$). When shifting to the global approach, an inverse relationship was observed.

The raw Global Ricci score (ρ) resulted in a statistically significant inverse correlation ($p < 0.01$). This indicates that architectures exhibiting the most negative Global Ricci during the first 40 epochs achieved lower long-term accuracy. By applying Fisher’s z-transformation to normalize the signal and eliminate architectural depth biases (excluding the 4-layer networks), this inverse correlation reached a Pearson r of -0.4867 ($p < 0.001$).

As shown in the parameter sweep, the inverse correlation of the normalized Global Ricci metric is maintained across the tested k -NN scales. After normalizing the eligible architectures for depth, the results indicate that architectures with the strongest early decrease in total graph distance tended to achieve lower long-term accuracy on CIFAR-10.

5.3 Summary of Key Findings

The experimental results presented in this section provide strong evidence supporting the “Deep Learning as Ricci Flow” hypothesis. The key findings can be summarized as follows:

1. **Ricci flow behavior is ubiquitous:** All tested architectures exhibit geometric transformations consistent with discrete Ricci flow, confirming that DNNs systematically simplify data geometry in a curvature-driven manner.
2. **Depth enhances geometric flow:** Deeper networks (11 layers) demonstrate stronger Ricci flow-like behavior compared to shallower ones (5 layers), with statistically significant correlations between geometric characteristics and classification accuracy (Pearson $r = -0.709$ for 11-layer vs. $r = -0.501$ for 5-layer networks).
3. **Architecture matters:** Wide architectures achieve the highest accuracy (98.76%), while bottleneck architectures show competitive performance despite fewer layers, suggesting that both network capacity and information compression contribute to effective data manifold simplification.
4. **Task difficulty influences performance:** Perceptually distinct classes (e.g., digits 1 vs 7, achieving 99.07% accuracy) are classified more accurately than perceptually similar classes (e.g., Shirts vs Coats, achieving 98.53%), reflecting the varying geometric complexity of different classification problems.
5. **Results are parameter-robust:** The observed relationships hold across different k -NN neighborhood sizes, with $k = 500$ consistently yielding optimal results for real-world image datasets.
6. **Curvature evolution is layer-dependent:** Layer-wise analysis of local Ricci coefficients reveals that geometric transformations are not uniformly distributed

across the network. Different layers exhibit distinct curvature trajectories over training, suggesting that the Ricci flow analogy holds at a local, per-layer level and that individual layers contribute differently to the overall geometric simplification of the data manifold.

7. **Curvature dynamics offer a complementary early stopping signal:** Tracking the evolution of Ricci coefficients over training epochs reveals a geometric stabilization pattern that aligns broadly with the convergence detected by standard validation-based early stopping. When configured appropriately, Ricci-based early stopping tends to identify convergence earlier than standard criteria, at the cost of a consistent but modest reduction in final model performance. These results suggest that curvature-based stopping captures meaningful convergence information, though it currently functions as a complementary signal rather than a standalone replacement, trading a degree of accuracy for earlier termination of training.
8. **Layer position affects the interpretation of Ricci values:** Filtering methods such as IQR or MAD can remove extreme Ricci coefficients, but they do not consider whether a removed value comes from an early, middle, or output-adjacent layer. Since the coefficients are generated by ordered layer transitions, a repeated change near the end of the network may reflect a systematic output-layer compression effect rather than random noise. Therefore, Ricci values should be interpreted together with their layer positions, not only as an unordered set of scalar observations.
9. **Curvature stabilization works as an early architectural filter:** A two-phase screening approach shows that Ricci-based early stopping triggers significantly earlier than standard methods (an average of 4.3 epochs compared to 16.8). While this early signal reflects a substantial part of the network’s potential, it can stop training too early for bottleneck architectures. This suggests it is most effective as an early screening tool rather than a precise stopping criterion.
10. **The predictive capacity of early geometric flow is dataset-dependent and scale-dependent:** In simpler classification tasks like Fashion-MNIST, early local Ricci scores show a moderate positive correlation (Pearson $r = +0.505$) with long-term accuracy, acting as a coarse architectural filter. However, on more complex datasets like CIFAR-10, the local signal becomes noisy and insignificant. Applying a global, depth-normalized approach via Fisher’s z-transformation on CIFAR-10 reveals an inverse relationship (Pearson $r = -0.4867$).

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